



Price Range Prediction of Accommodations around Mae Fah Luang University Campus Using Machine Learning

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BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING

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UNIVERSITY
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THIS COMPUTER ENGINEERING PROJECT HAS BEEN
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EXAMINING COMMITTEE



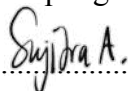
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Abstract

Finding affordable and suitable housing is a major concern for many people, particularly students who often rely on dormitories for their accommodation needs. Predicting the expected price range of dormitories based on relevant features can help people make informed decisions about housing and improve their financial well-being. In this project, we develop machine learning models to predict the expected price range of dormitories based on various features, such as location, room size, and amenities, as well as additional factors like proximity to the university, accessibility to transportation, and student demographics.

We utilize historical data for both training and testing our models, assessing their performance through a range of metrics encompassing accuracy, precision, recall, and F1 score. Our findings indicate the proficient capability of our models in accurately predicting the anticipated price range of dormitories. This insight proves valuable for both renters and dormitory owners, as it enables informed decision-making regarding housing. The success of our machine learning techniques in forecasting dormitory prices carries significant implications, potentially contributing to the enhancement of housing affordability and accessibility issues of paramount importance to many individuals.

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CHAPTER 1

INTRODUCTION

1.1 Background and Rational

Finding an affordable and suitable dormitory is crucial for students and people who want to save on housing costs. Predicting the expected price range of dormitories based on relevant features such as location, room size, amenities, and other factors like proximity to the university, accessibility to transportation, and student demographics, can help people make informed decisions. Machine learning techniques can be used to develop models that predict the expected price range of dormitories based on historical data and current market trends. These models can help people find the best housing options that fit their budget and preferences, while also providing dormitory owners with insights into setting competitive prices. As such, developing accurate dormitory expected price range prediction models can benefit both renters and dormitory owners.

1.2 Objective

- 1.2.1 To create a machine learning model that can predict the price ranges of accommodation based on the user's preferences of the facility.
- 1.2.2 To develop a web application that offers users an estimated price range for rental accommodations based on their input preferences.

1.3 Scope

- 1.3.1 This web application is used for predicting the price ranges of accommodations around Mae Fah Luang University areas.
- 1.3.2 The dataset contains accommodation information gathered from around Mae Fah Luang University areas, wherein survey participants scored collected data based on their satisfaction with certain features.

1.4 Methodology

- 1.4.1 Find the topic of the project and present it to the advisor and committee.
- 1.4.2 Explore the dataset.
- 1.4.3 Dataset preprocessing..
- 1.4.4 Model training
- 1.4.5 Model testing and Evaluate result
- 1.4.6 Model Selection
- 1.4.7 Pre-project presentation
- 1.4.8 Design Web Application
- 1.4.9 Website Development.
- 1.4.10 Combine model to the web application
- 1.4.11 System Testing
- 1.4.12 Final project presentation

1.5 Plan

	Pre-Project				Project				
Plan/month	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Find the topic of the project and present it to the advisor and committee.									
Explore the dataset.									
Dataset preprocessing..									
Model training									
Model testing and Evaluate result									
Model Selection									
Preproject presentation									
Design Web Application									
Website Development.									
Combine model to the web application									
System Testing									
Final project presentation									

Table 1.1 Plan

1.6 Expected Result

- 1.6.1 The Machine learning model can predict the price ranges of accommodation based on the user's preferences of the facility.
- 1.6.2 The web application can display the expected results of estimated price range for rental accommodations based on the user's input preferences.

1.7 Place and Equipment

- 1.7.1 Dormitory
- 1.7.2 Laptop or PC desktop
- 1.7.3 Python (Machine learning)
- 1.7.4 HTML,CSS,PHP,JavaScript,SQL (web application development)

CHAPTER 2

Literature Review

2.1 Related Theory

2.1.1 Web Application

A web application (web app) is an application program stored on a remote server, accessed through a browser interface over the internet. It encompasses various functionalities and services delivered through web interfaces. These applications offer dynamic and interactive features to users. In the development of our project, several tools and frameworks were instrumental. Specifically, we employed Frameworks such as Bootstrap, jQuery 3.6.0, Sweetalert, boxicon, and Flask. Each tool served a specific purpose, facilitating the creation of an engaging and functional web interface to predict accommodation price ranges and enable data collection.[1]

2.2 Related Research

2.2.1 Factors Influencing the Selection of Private Dormitory for Mae Fah Luang University Students[2] conducted a study to identify the factors that influence the selection of private dormitories near Mae Fah Luang University in Chiang Rai. The researchers identified the main factor affecting dormitory selection to be the personnel factor. The study found that the opinion level of students was very important, with the four most important factors being products (room), price, location, and physical aspects. The data set was collected through a Google Form questionnaire and analyzed for percentage, mean, standard deviation, and graph data. The study showed that most respondents were female students in their fourth year of university education, with an average income and expenditure in the range of 5,001-10,000 baht. The majority of students had been informed about dormitory locations near the university before deciding to rent, and desired a room with complete facilities within 1-2 kilometers of the university.

2.2.2 Modeling of apartment prices in a Colombian context from a machine learning approach with stable-important attributes[3]. conducted a study about develop a machine learning model for online pricing of apartments in Colombia, comparing the predictive capacity of various models, studying the effect of text attributes on the predictive capability, and identifying stable-important attributes for inferential interpretation. The sample consisted of 15,177 observations of real estate. The methods of assembly (random forest and bagging) showed predictive superiority over others, with the BAG method achieving the best results. Four primary attributes (area.build.m2, bathrooms, pre.m2.mean.zone, and strat.rec) were found to provide most of the information needed for predictions. Although some text attributes showed a statistically significant relationship with the property price, their contribution to the predictive capacity was minimal. The stability of the top 4 attributes was confirmed, and the important roles of two text-derived attributes (stim.buy and longi) were highlighted.

2.3 Related Technology:

2.3.1 Machine Learning:

Machine learning is a subfield of artificial intelligence that uses algorithms and statistical models to enable computer systems to learn and improve from experience [4]. The goal of machine learning is to create models that can recognize meaningful patterns in data, and use those patterns to make predictions or decisions. Machine learning is used in many applications, including image recognition, natural language processing, fraud detection, and personalized recommendations, among others. Popular machine learning algorithms include neural networks, decision trees, random forests, and support vector machines, making it a powerful tool for automating complex tasks and improving decision-making [5][6].

2.3.2 Machine Learning Model

A machine learning model is a program designed to detect patterns or make predictions based on new and unseen data. In the context of predicting accommodation price ranges, machine learning models can process data encoding, perform resampling techniques to address data imbalance, use sequential features selection to optimize relevant features, and undergo hyperparameter tuning to enhance their predictive accuracy. Through training on extensive datasets, these models learn to recognize critical patterns or generate specific outputs pertinent to the given task. The resulting output is often a computer program equipped with predefined rules and data structures, referred to as a machine learning model. [7]

2.3.2.1 Logistic Regression

Logistic Regression stands out as a model frequently used in predicting accommodation price ranges. Logistic Regression models the probability of a categorical outcome, making it suitable for binary classification tasks. It's particularly effective in scenarios where understanding the relationship between the dependent and independent variables is crucial. These models leverage data encoding techniques, perform resampling methods to tackle data imbalance issues, employ sequential features selection to optimize relevant features, and undergo hyperparameter tuning to enhance their predictive accuracy. Through training on extensive datasets, Logistic Regression learns to recognize critical patterns or generate specific outputs pertinent to the given task. The resulting output is often a computer program equipped with predefined rules and data structures, referred to as a machine learning model. [8]

2.3.2.2 Random Forest

Random Forest, an ensemble learning technique, is extensively utilized in predicting accommodation price ranges. It constructs multiple decision trees and combines their predictions to improve accuracy and reduce overfitting. Random Forest operates by generating a multitude of decision trees during training and outputs the mode of the classes (classification) or the mean prediction (regression) of the individual trees. In the context of accommodation price prediction, it excels in handling large datasets with many features, offering high accuracy and robustness. Similar to other machine learning models, Random Forest benefits from data encoding techniques, resampling methods to handle data imbalance, sequential features selection, and hyperparameter tuning to fine-tune its predictive capabilities. This iterative learning process enables Random Forest to discern critical patterns, enhancing its predictive prowess in determining accommodation price ranges. [9]

2.3.2.3 Decision Trees

Another fundamental algorithm in predicting accommodation price ranges is the Decision Trees model. Decision Trees are recursive partitioning structures that segment the data into subsets based on various attributes. They make decisions by traversing the tree from the root to the leaf nodes, where each node represents a feature and each branch a decision rule leading to the next node. Decision Trees are valued for their interpretability and ease of understanding, making them insightful for uncovering key factors influencing accommodation price ranges. Like other machine learning models, Decision Trees benefit from data encoding techniques, resampling methods to address data imbalance, sequential features selection, and hyperparameter tuning, all contributing to optimizing their predictive accuracy. By scrutinizing extensive datasets, Decision Trees discern crucial patterns, contributing significantly to predicting accommodation price ranges. [10]

2.3.3 Classification:

In machine learning, classification is a type of supervised learning used to predict the category of a given data point. This involves assigning data points to one or more target labels, categories or classes. For instance, a spam detection algorithm may classify emails as either "spam" or "not spam." [11]

Classification algorithms are:

- 1.K-nearest neighbor
- 2.Decision trees
- 3.Naive Bayes
- 4.Artificial neural networks
- 5.Logistic Regression
- 6.Support Vector Machine

Figure 2.1 shows machine learning classification algorithms

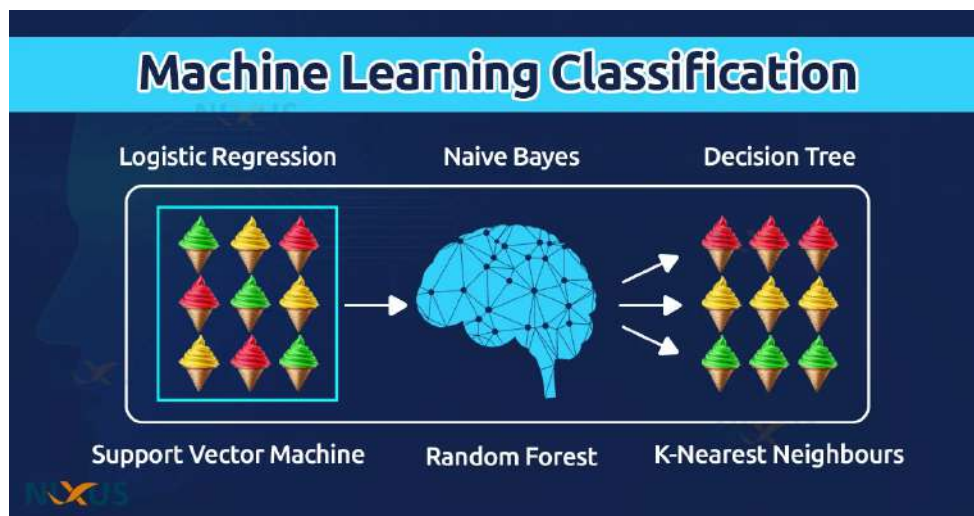


Figure 2.1 Classification Picture[12]

From: <https://nixustechnologies.com/machine-learning-classification/>

CHAPTER 3

METHODOLOGY

3.1 Workflow

The overview of the project. Figure 3.1 shows the overview workflow

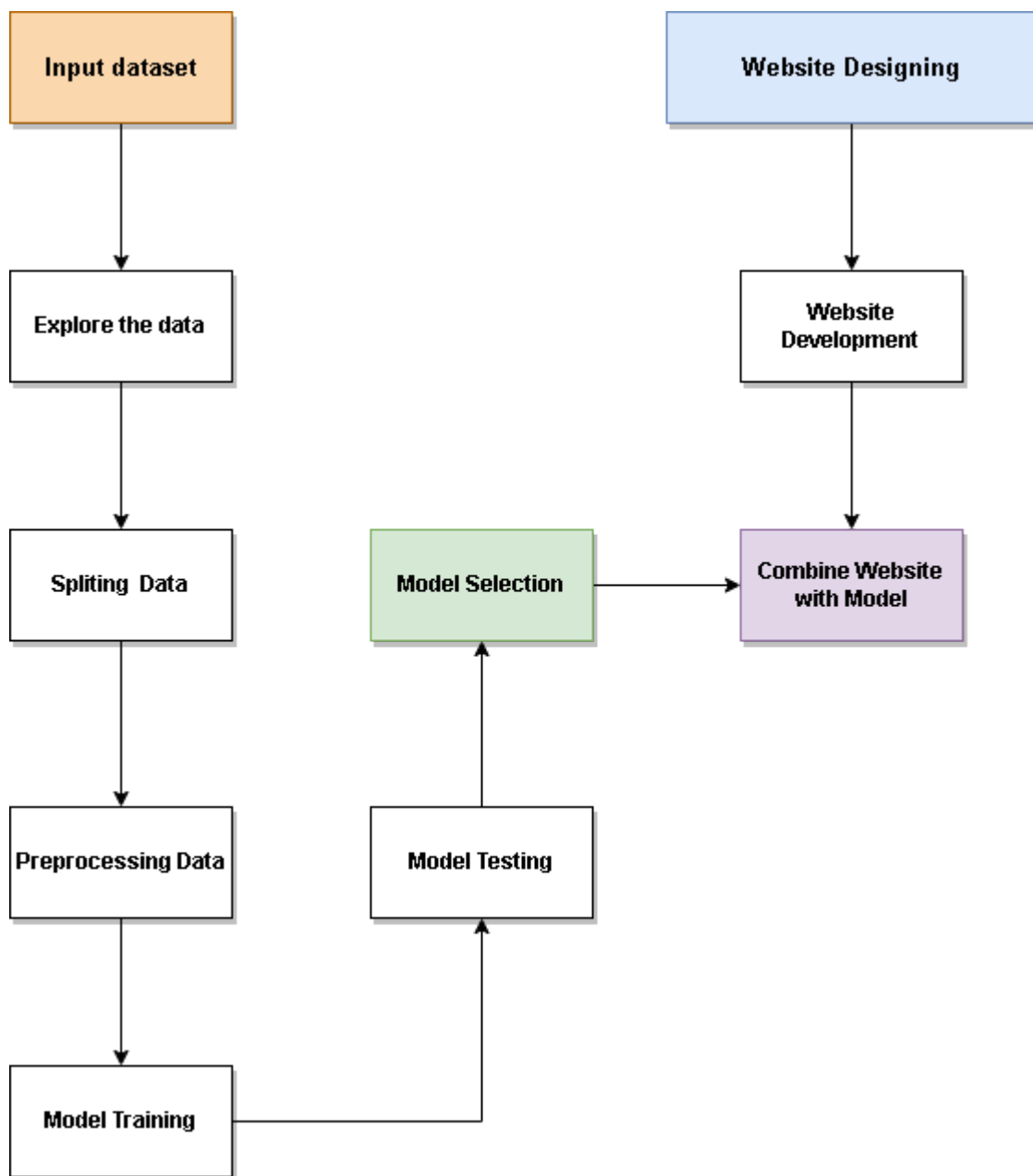


Figure 3.1 Overview Workflow

3.2 Data description

No.	Features Name	Description	Features Class
1	sex	sex of student	Men,Women,LGBTQ+
2	age	age of student	น้อยกว่า 18,19 - 21,22 - 24,มากกว่า 24
3	education_level	education level of student	ปี 1,ปี 2,ปี 3,ปี 4,มากกว่าปี 4
4	school	like a major that student study	สำนักวิชาการแพทย์บูรณาการ,สำนักวิชาการจัดการ,สำนักวิชาพยาบาลศาสตร์,สำนักวิชาเวชศาสตร์ชะลอวัยและฟื้นฟูสุขภาพ,สำนักวิชาเทคโนโลยีสารสนเทศ,สำนักวิชาวิทยาศาสตร์สุขภาพ,สำนักวิชานิติศาสตร์,สำนักวิชาวิทยาศาสตร์เครื่องสำอาง,สำนักวิชาเงินวิทยา,สำนักวิชานวัตกรรมสังคม,สำนักวิชาทันตแพทยศาสตร์,สำนักวิชาศิลปศาสตร์,สำนักวิชาอุตสาหกรรมเกษตร,สำนักวิชาวิทยาศาสตร์,สำนักวิชาแพทยศาสตร์
5	average_monthly_income	average income that student got in each month	น้อยกว่า 1,000 ,1,000 - 3,000, 3,001 - 5,000,5,001 - 10,000,มากกว่า 10,000
6	average_monthly_expenditure	average expend that student got in each month	น้อยกว่า 1,000 ,1,000 - 3,000, 3,001 - 5,000,5,001 - 10,000,มากกว่า 10,000
7	vehicle_possession	vehicle that student have	ไม่มี,จักรยานยนต์ (มอเตอร์ไซค์),รถยนต์ส่วนตัว,จักรยาน
8	travel_to_university_method	the way that student use to go to university	จักรยานยนต์ (มอเตอร์ไซค์),รถยนต์ส่วนตัว,รถเข็น,เดิน,จักรยาน

No.	Features Name	Description	Features Class
9	domicile	domicile of student	ภาคเหนือ,ภาคกลาง,ภาคตะวันออกเฉียง,ภาคใต้
10	security_deposit_price	security deposit price of accommodation	น้อยกว่า 2,000, 2,001 - 5,000 , 5,001 - 8,000 , มากกว่า 8,001
11	dormitory_location	accommodation location	ใกล้มหาวิทยาลัย', 'ใกล้ร้านอาหาร', 'ใกล้เซเว่น', 'ใกล้สถานที่ท่องเที่ยว', 'อื่นๆ'
12	gained_info_by	gained info of accommodation by	ติดต่อด้วยตัวเอง', 'เพื่อน', 'โซเชียลมีเดีย', 'เจ้าของหอพัก', 'website', 'อื่นๆ'
13	facilities	facilities of accommodation	ห้องเปล่าไม่มีเฟอร์นิเจอร์ (Empty room)', 'ห้องมีสิ่งอำนวยความสะดวกบางส่วน (Semi-Furnished) (ควรมีแอร์, ที่วี, โต๊ะเครื่องแป้ง, เตียง, ตู้เย็น, เครื่องทำน้ำอุ่น, ระเบียงห้อง เป็นต้น)', 'ห้องมีสิ่งอำนวยความสะดวกครบ (Fully-Furnished)(ควรมีโซนห้องครัว, ห้องนั่งเล่น, Wi-Fiส่วนตัว เป็นต้น)'
14	travel_time	time that student use to go to university	น้อยกว่า 5 นาที', '6 - 10 นาที', '10 - 15 นาที', '16 - 20 นาที', 'มากกว่า 20 นาที'
15	distance_to_university	distance from accommodation to university	น้อยกว่า 1 กิโลเมตร', '1 - 2 กิโลเมตร', '2 - 3 กิโลเมตร', '3 - 4 กิโลเมตร', '4 - 5 กิโลเมตร', 'มากกว่า 5 กิโลเมตร'
16	room_size	room size of accommodation (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'

No.	Features Name	Description	Features Class
17	cleanliness	how clean of accommodation (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
18	amenities	amenities in accommodation (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
19	decoration	decoration in accommodation (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
20	balcony_size	balcony size in accommodation (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
21	price_rate	price rate of accommodation (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
22	utility_bills	utility bills from accommodation (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
23	security_deposit_price_rate	security deposit price rate of accommodation (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
24	dormitory_common_fee	dormitory common fee (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
25	located_near_a_convenience_store	how close of accommodation and store (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
26	located_at_a_far_from_the_university	how far of accommodation and university (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
27	polite_speaking_staff	staff have a polite conversation (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
28	staff_solved_the_problem_well	staff can solved problem as well (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'

No.	Features Name	Description	Features Class
29	dormitory_environment	environment of accommodation (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
30	enough_parking_area	parking of accommodation enough for student (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
31	high_security	accommodation have a high security (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
32	round_the_dormitory_is_clean	around of accommodation area is clean (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
33	various_services_in_the_dormitory	service of accommodation (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
34	Other_tenants_have_good_manners	other living have a good manner (Satisfaction)	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'

Table 3.1 Data features

3.3 Data Exploration

The dataset used in this project, collected in the project “Factors Influencing the Selection of Private Dormitory for Mae Fah Luang University Students” by (Yodsarn et al.2020), consists of 382 records and 35 features., with four classes of labels. Through a survey of the data, it was determined that the dataset contains no missing values, providing a reliable foundation for machine learning analysis. the details show in table 3.1

3.4 Data Distribution

Exploratory data analysis was conducted using Seaborn [13], Matplotlib [14] and Google Sheets to visualize the distribution of various features. Histograms generated with Seaborn and Matplotlib provided insights into the frequency distribution of attribute values, highlighting central tendencies and potential outliers.

1. Label class distribution

The distribution of label classes in the dataset is depicted in the histogram below. The imbalance among the classes is evident, where Class 0 (Lower than 2,000) comprises 23 records, Class 1 (2,001 - 5,000) consists of 258 records, Class 2(5,001 - 8,000) has 88 records, and Class 3(More than 8,001) is represented by only 13 records. This illustration clearly demonstrates the significant class imbalance, with certain classes having notably fewer instances compared to others. Figure 3.2 shows the Label distribution

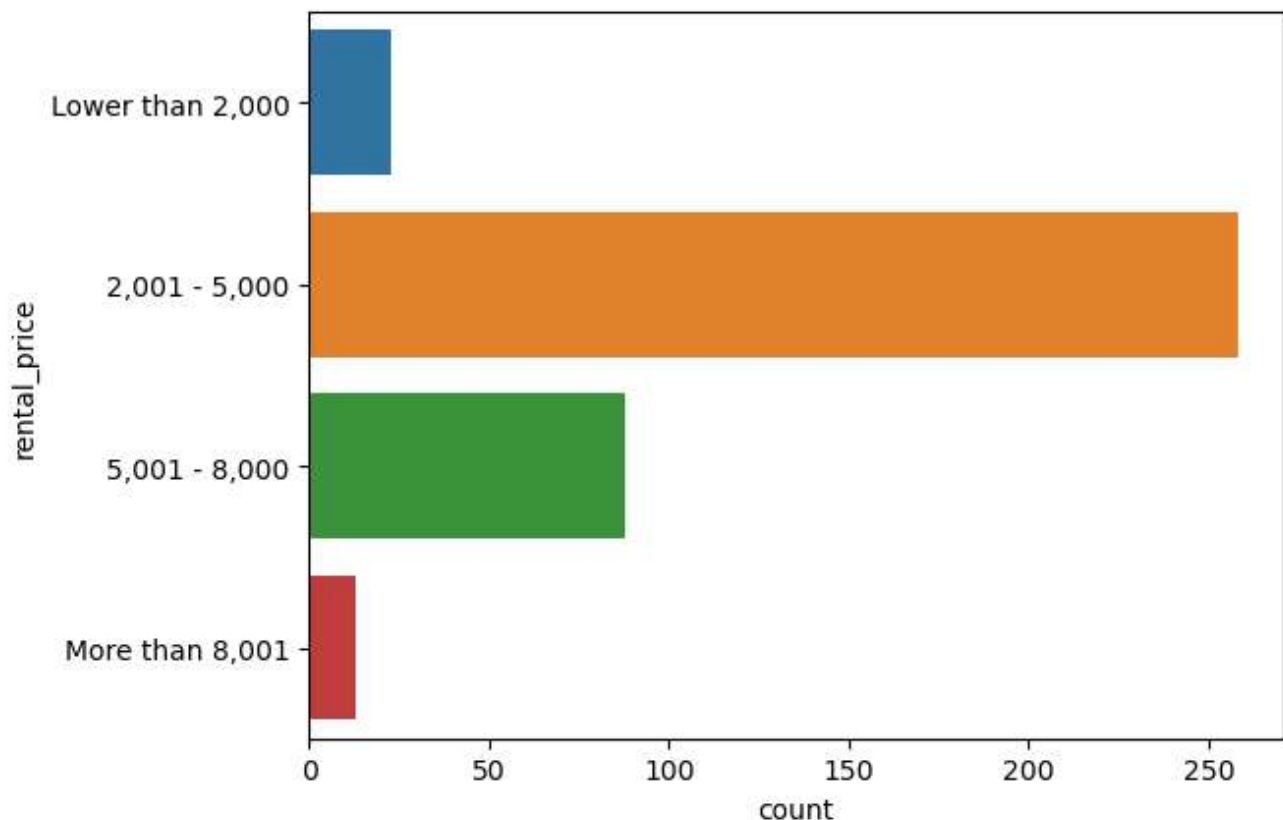


Figure 3.2 Label distribution

2. Features distribution

The distribution of 34 feature classes in the dataset is illustrated in the following figures: Figure 3.3 to Figure 3.36. These figures represent the distribution patterns of the respective features, providing insights into their frequency or occurrence within the dataset.

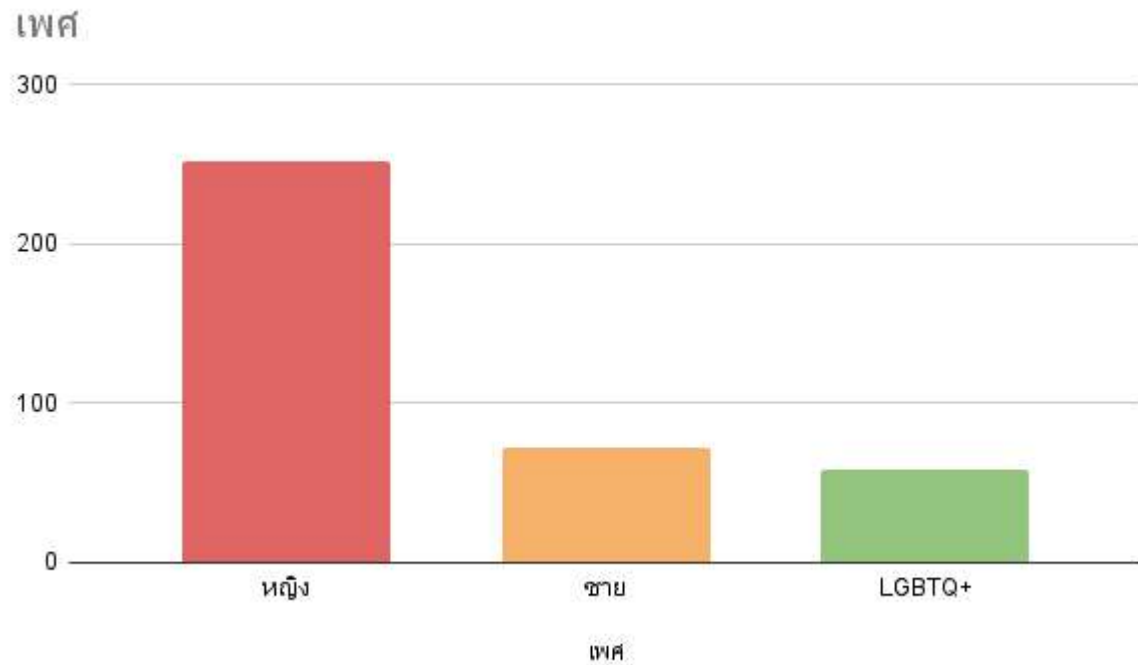


Figure 3.3 Sex distribution

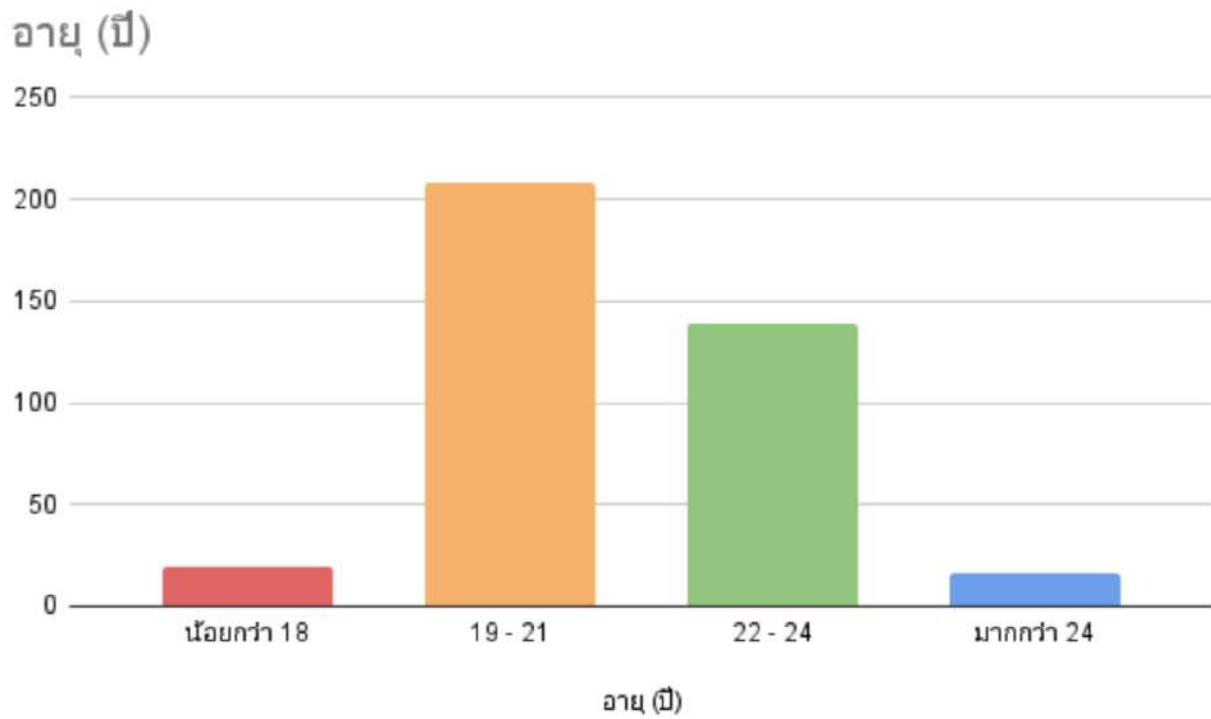


Figure 3.4 Age distribution

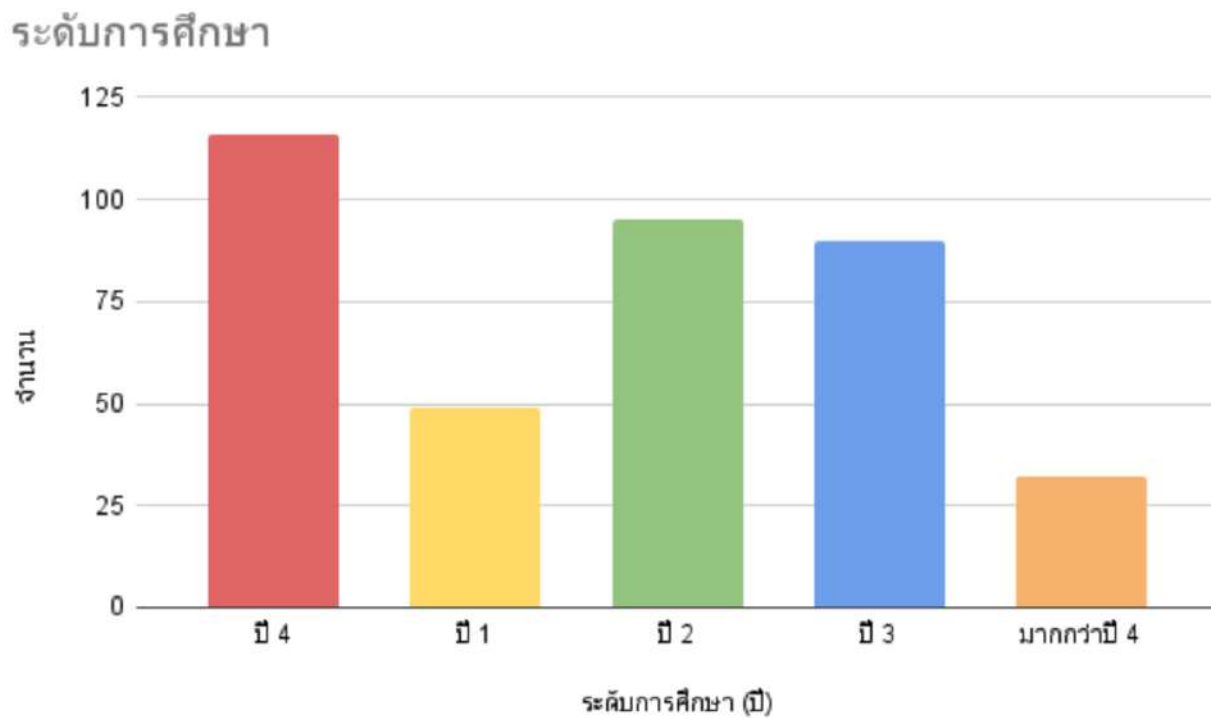


Figure 3.5 Education distribution

order from top to bottom:

1. สำนักวิทยบาลศาสตร์
2. สำนักวิชานิติศาสตร์
3. สำนักวิชาการจัดการ
4. สำนักวิชาจันวิทยา
5. สำนักวิชาวิทยาศาสตร์
6. สำนักวิชาวิทยาศาสตร์เครื่องสำอาง
7. สำนักวิชาเทคโนโลยีสารสนเทศ
8. สำนักวิชาการแพทย์บูรณาการ
9. สำนักวิชาศิลปศาสตร์
10. สำนักวิชาอุตสาหกรรมเกษตร
11. สำนักวิชานวัตกรรมสังคม
12. สำนักวิชาวิทยาศาสตร์สุขภาพ
13. สำนักวิชาทันตแพทยศาสตร์
14. สำนักวิชาเวชศาสตร์ชะลอวัยและฟื้นฟูสุขภาพ
15. สำนักวิชาแพทยศาสตร์

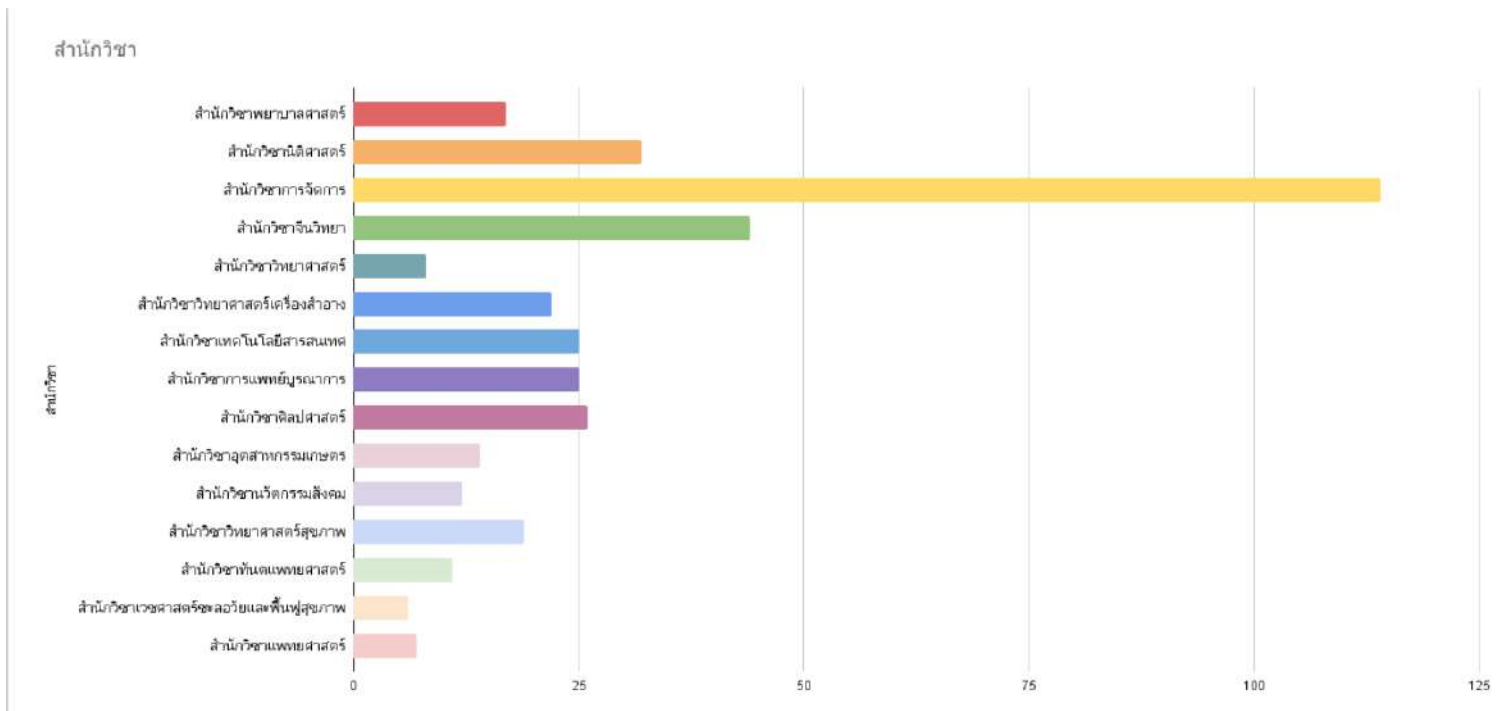


Figure 3.6 School distribution

รายได้เฉลี่ยต่อเดือน (บาท)

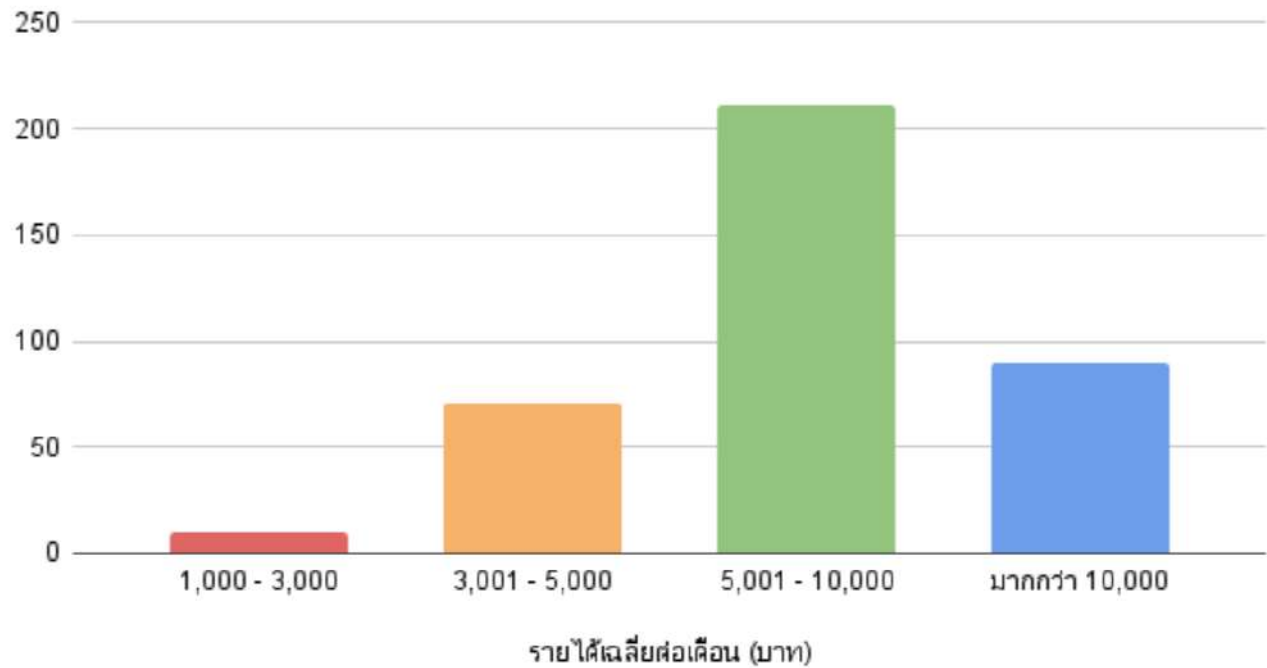


Figure 3.7 Average income distribution

รายจ่ายเฉลี่ยต่อเดือน (บาท)

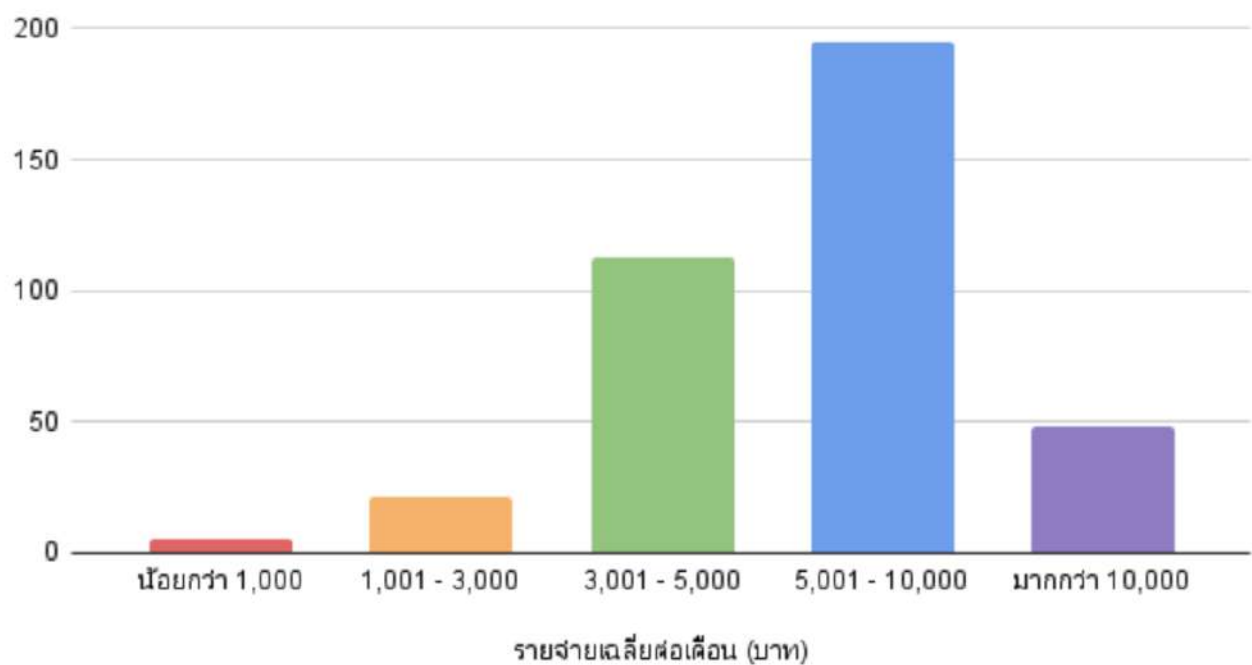


Figure 3.8 Average expenses distribution

การครอบครองยานพาหนะ

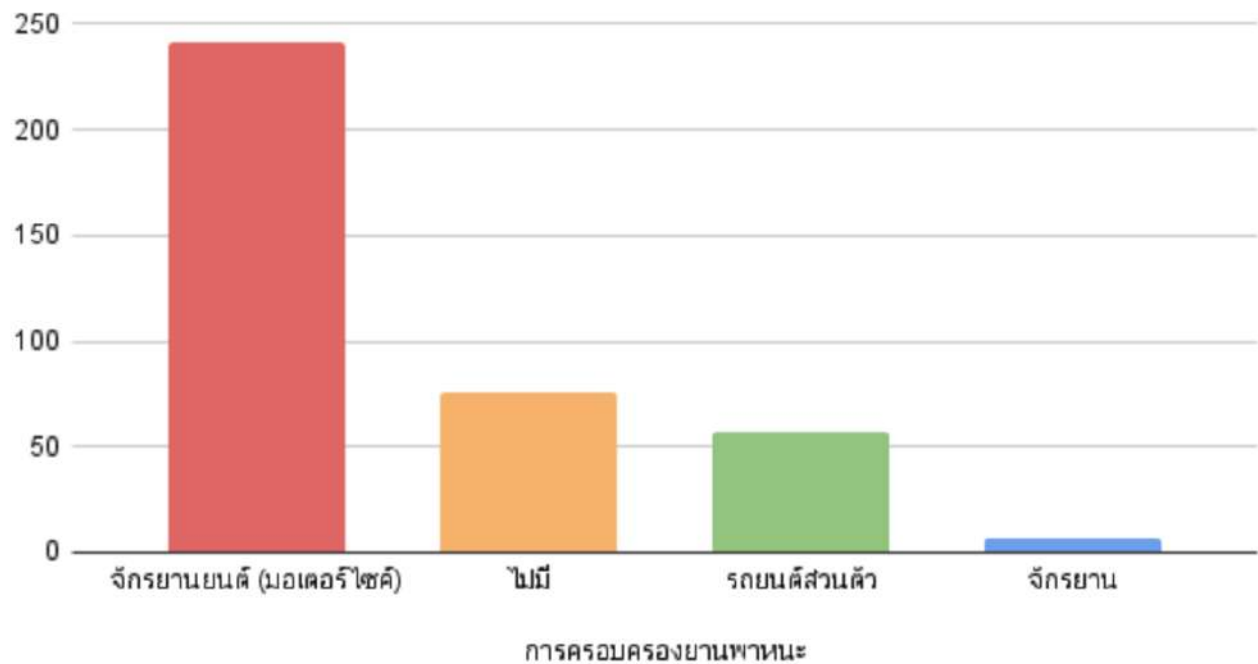


Figure 3.9 Vehicle possession distribution

วิธีการเดินทางไปมหาวิทยาลัย

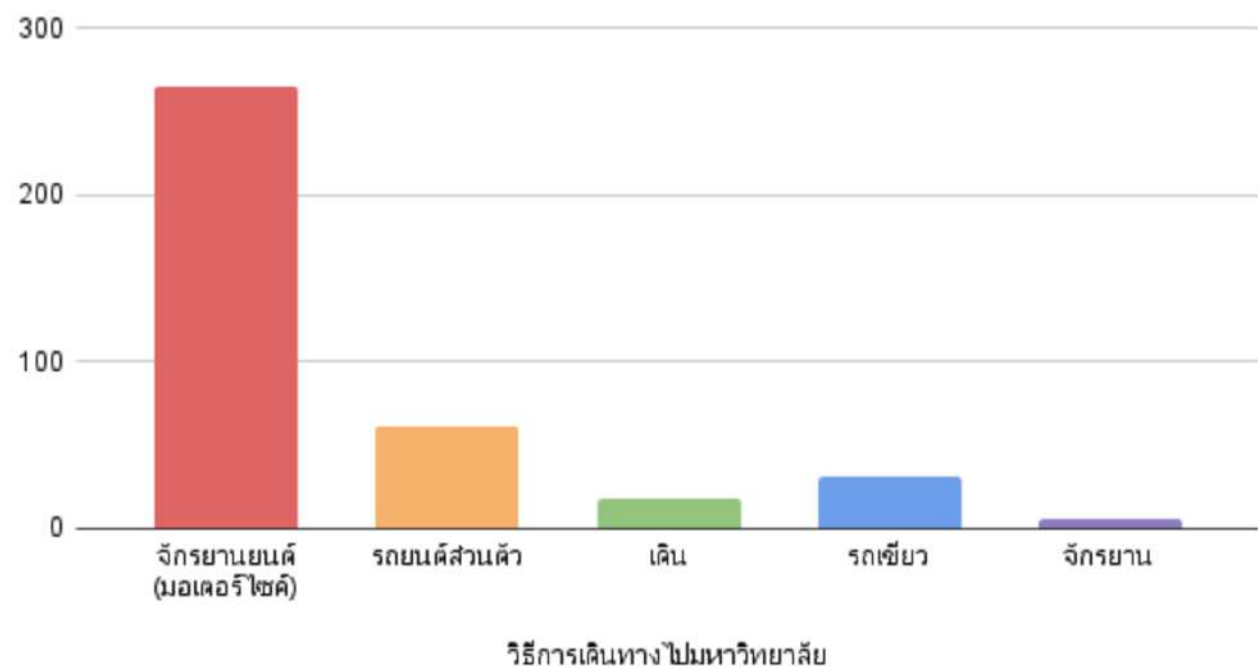


Figure 3.10 Travel method distribution

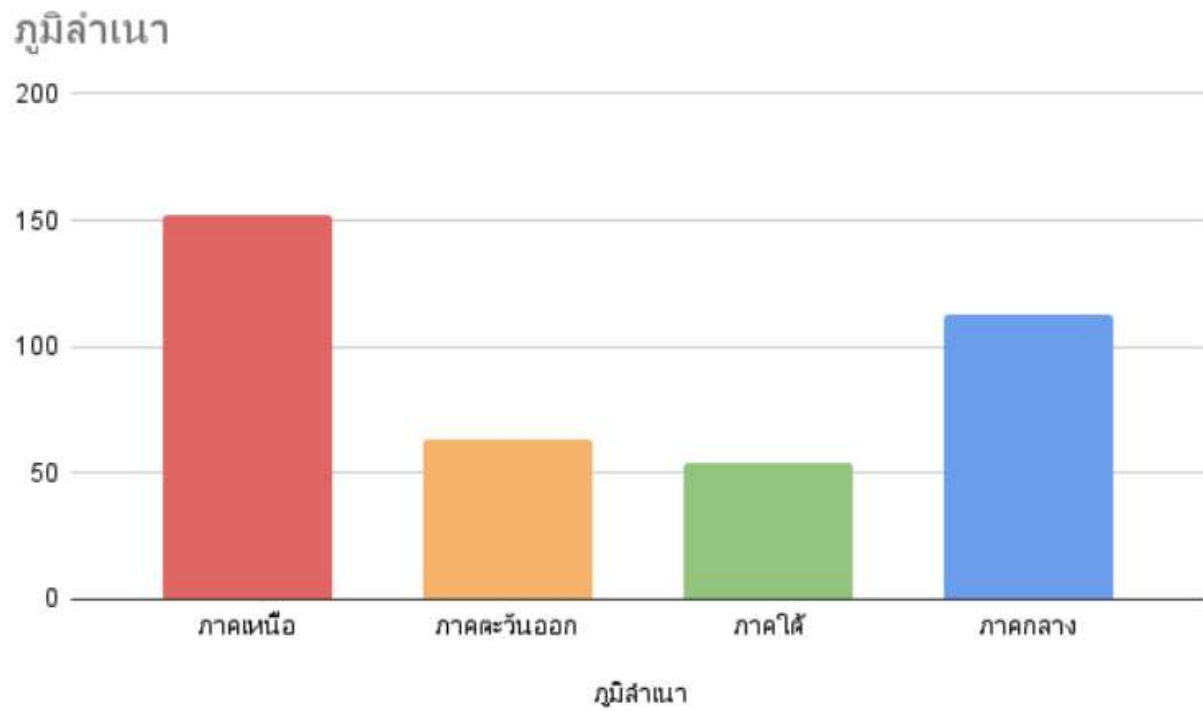


Figure 3.11 Domicile distribution

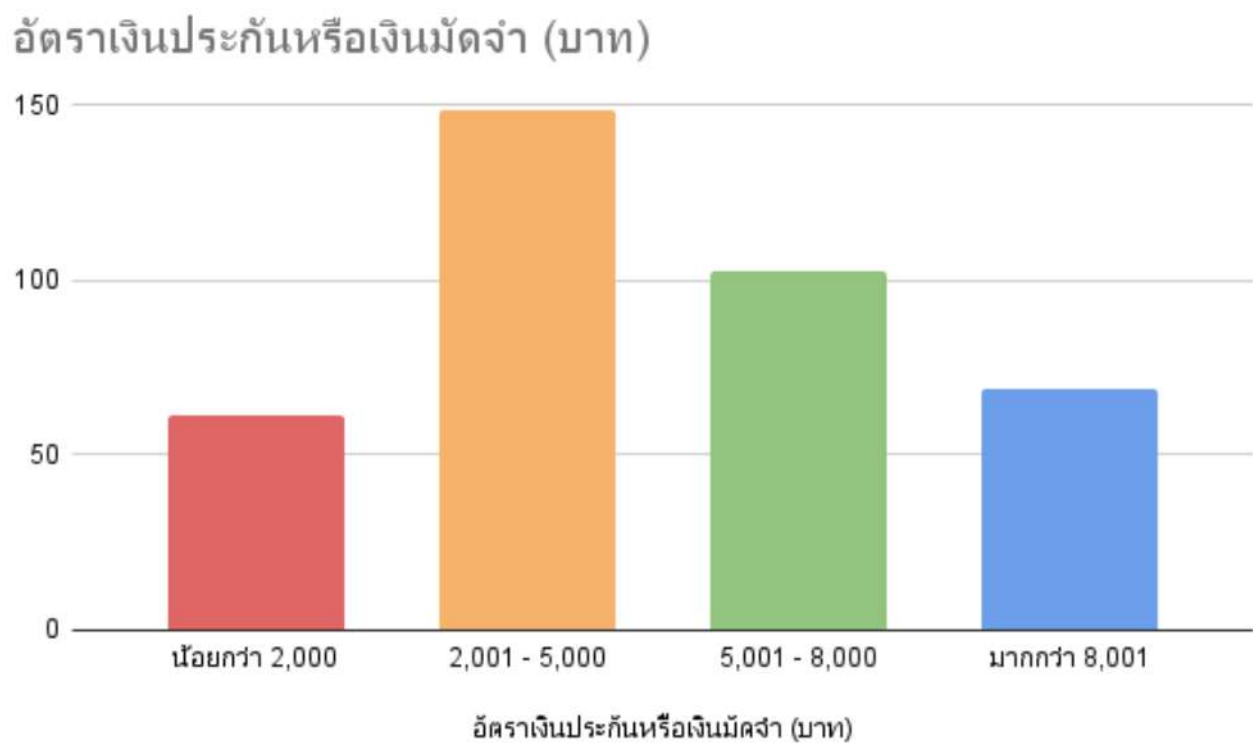


Figure 3.12 Security distribution

บริเวณที่ตั้งของห้องพักของท่าน ณ ปัจจุบัน

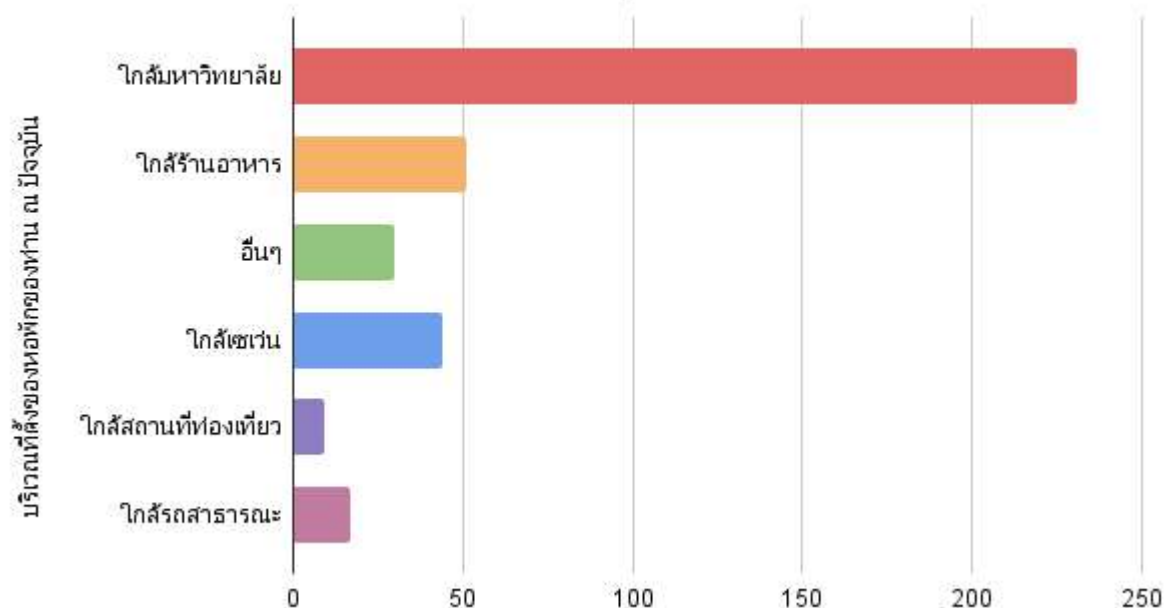
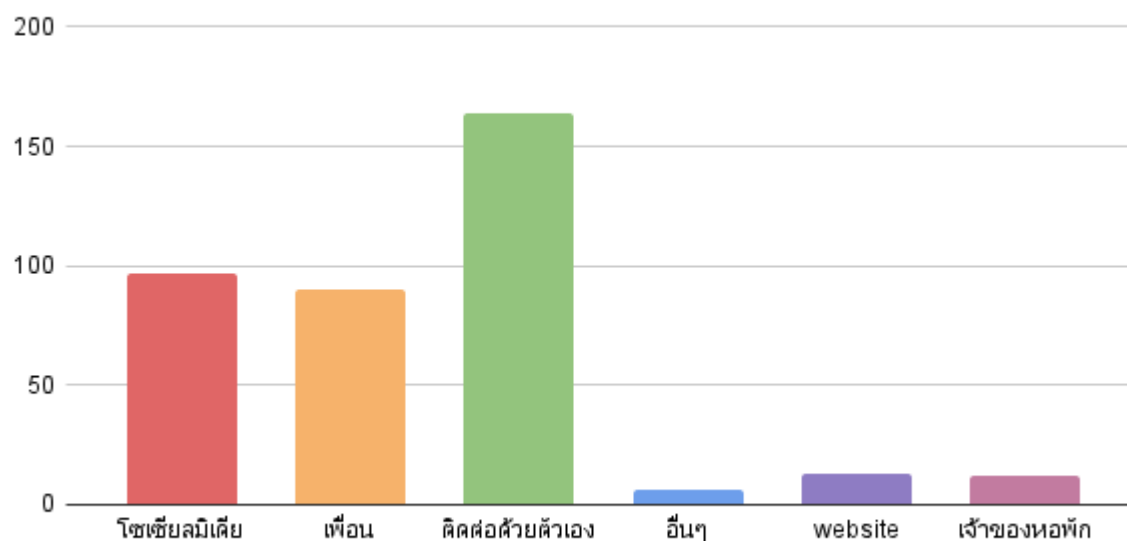


Figure 3.13 Accommodation location distribution

ท่านหาข้อมูลจากแหล่งข้อมูลใดก่อนการตัดสินใจเช่าห้องพัก ณ ปัจจุบัน



ท่านหาข้อมูลจากแหล่งข้อมูลใดก่อนการตัดสินใจเช่าห้องพัก ณ ปัจจุบัน

Figure 3.14 Gained info method distribution

- : Semi-Furnished
- : Fully-Furnished
- : Empty Room

สิ่งอำนวยความสะดวกภายในห้องพัก

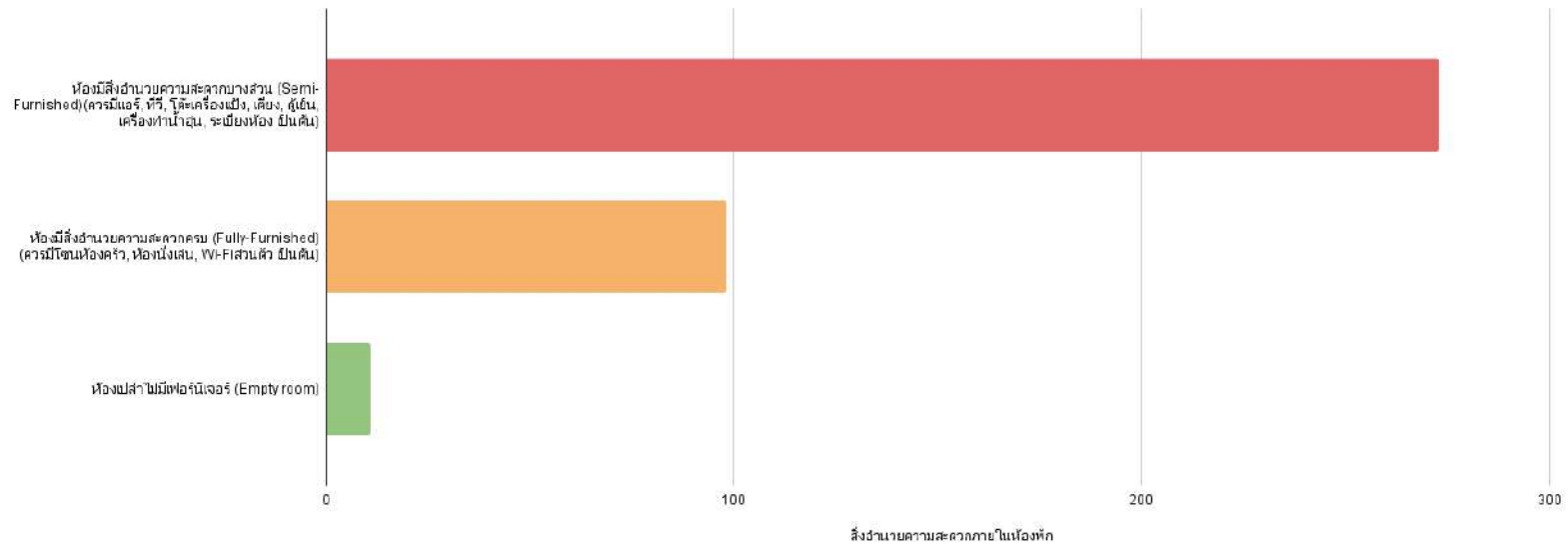


Figure 3.15 Facilities distribution

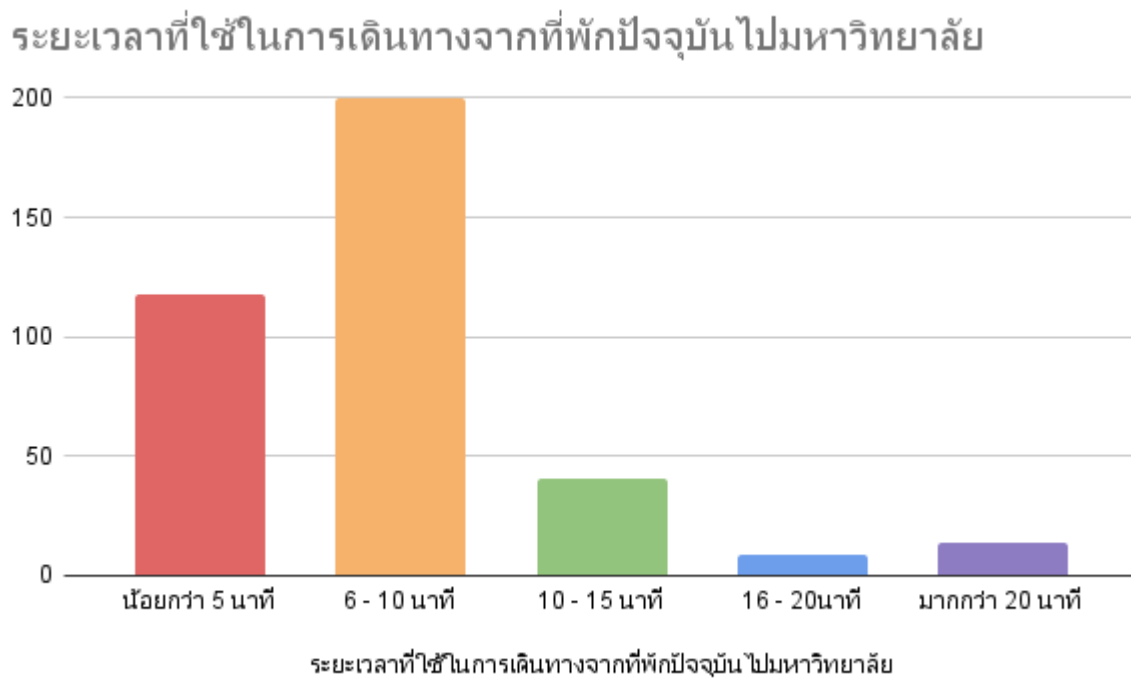


Figure 3.16 Travel time distribution

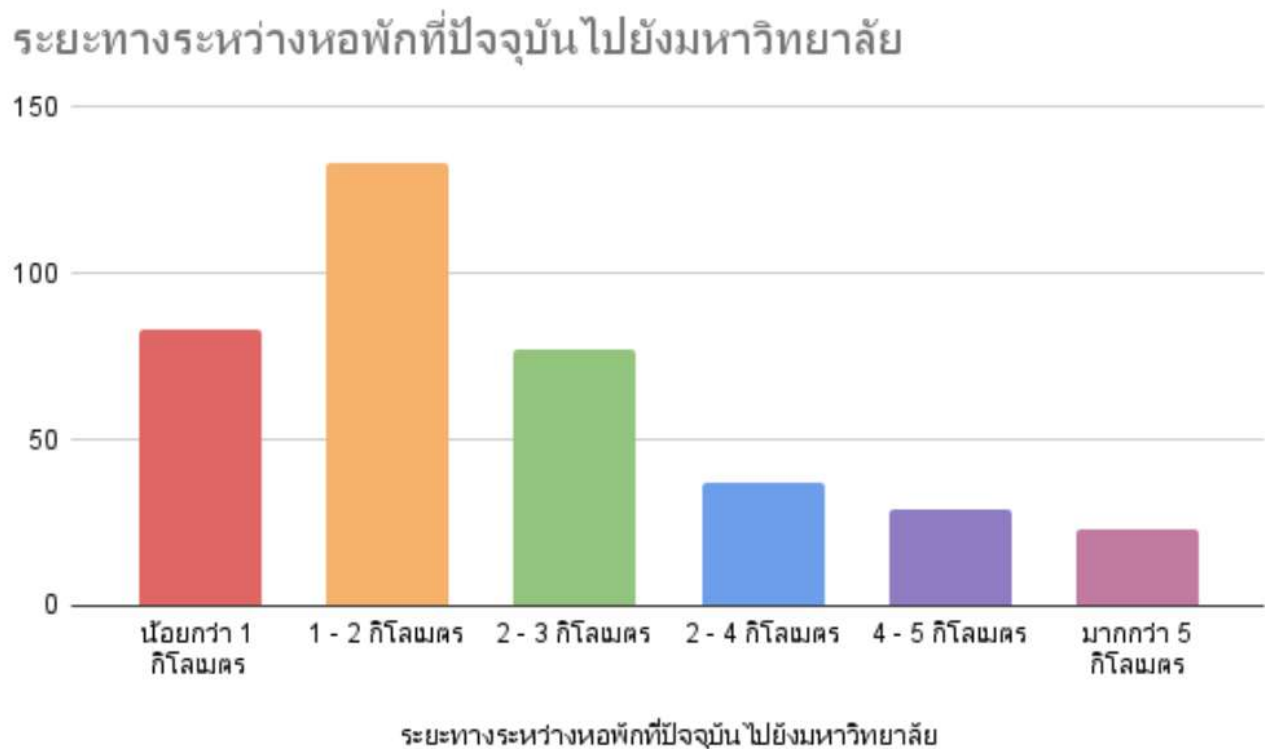


Figure 3.17 Distance to University distribution

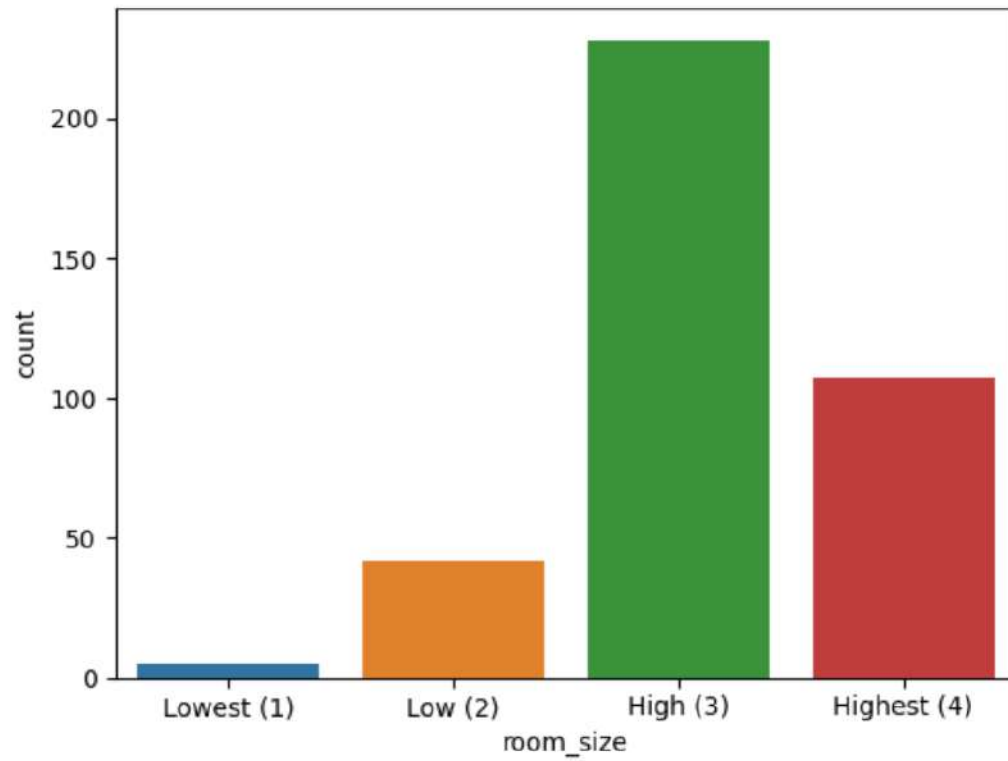


Figure 3.18 Domicile distribution

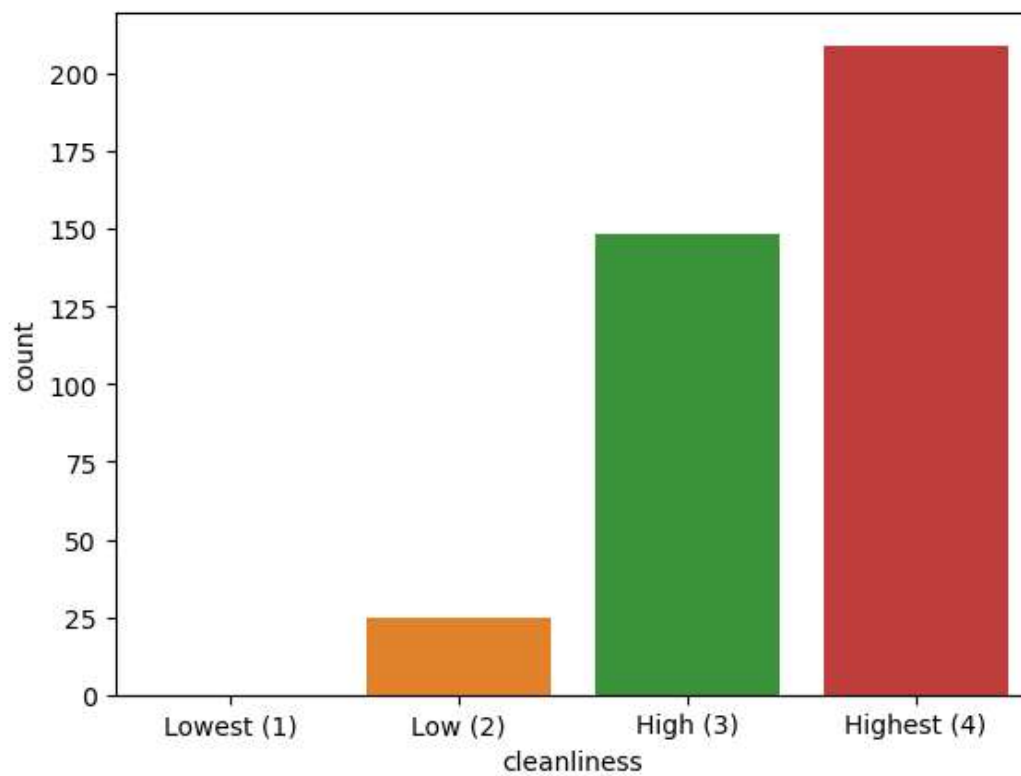


Figure 3.19 Cleanliness distribution

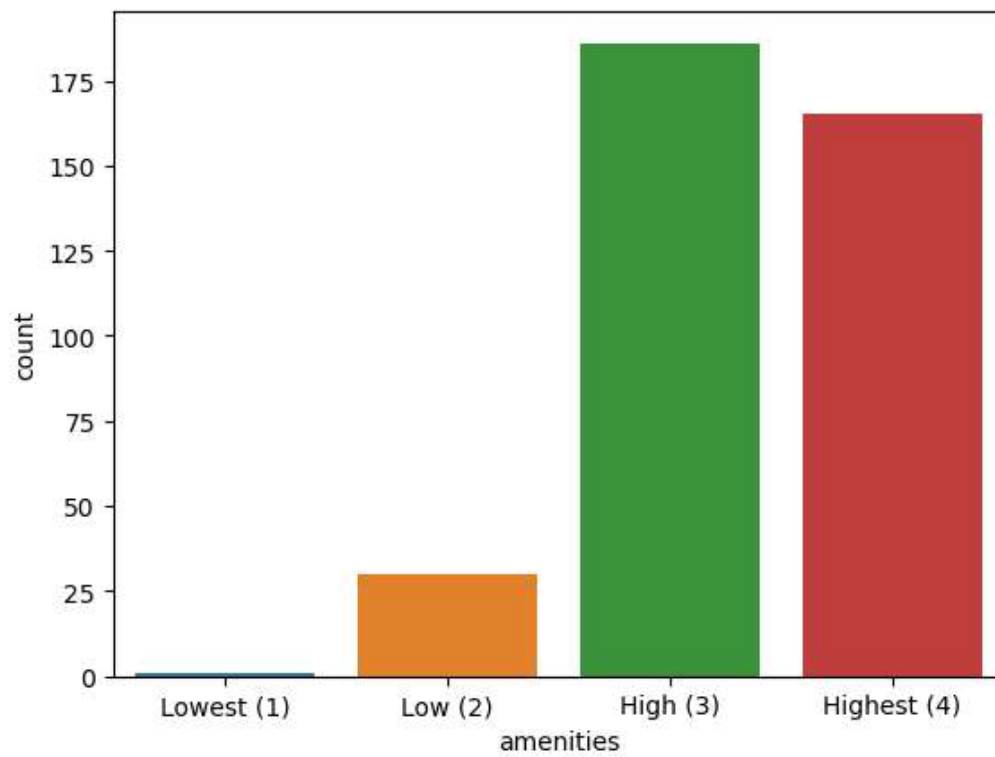


Figure 3.20 Amenities distribution

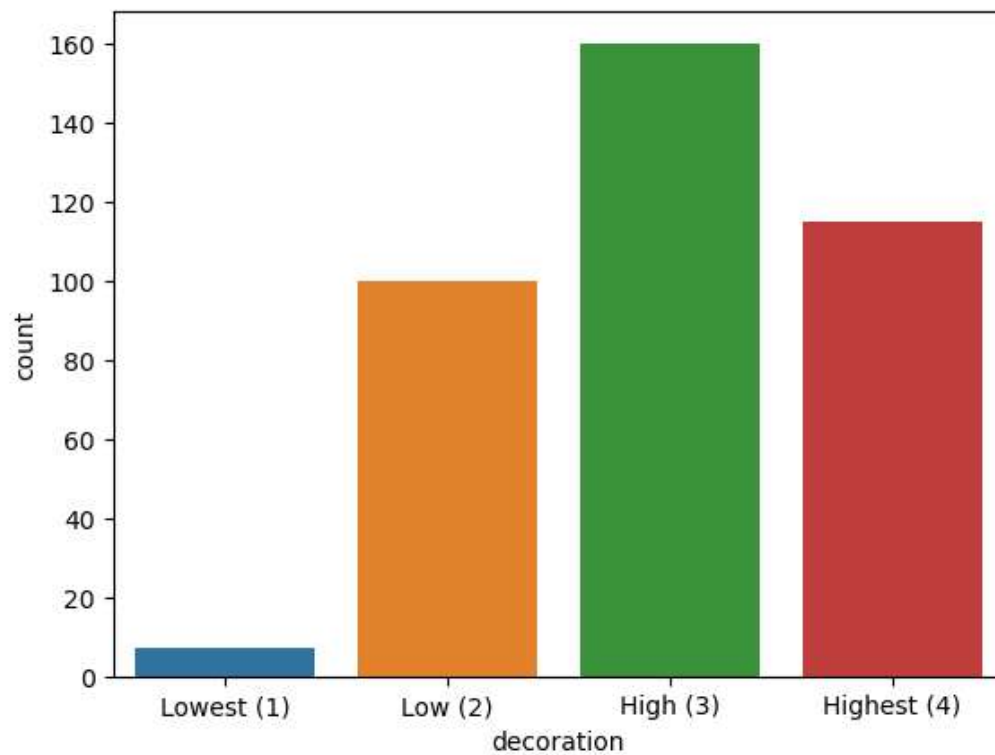


Figure 3.21 Domicile distribution

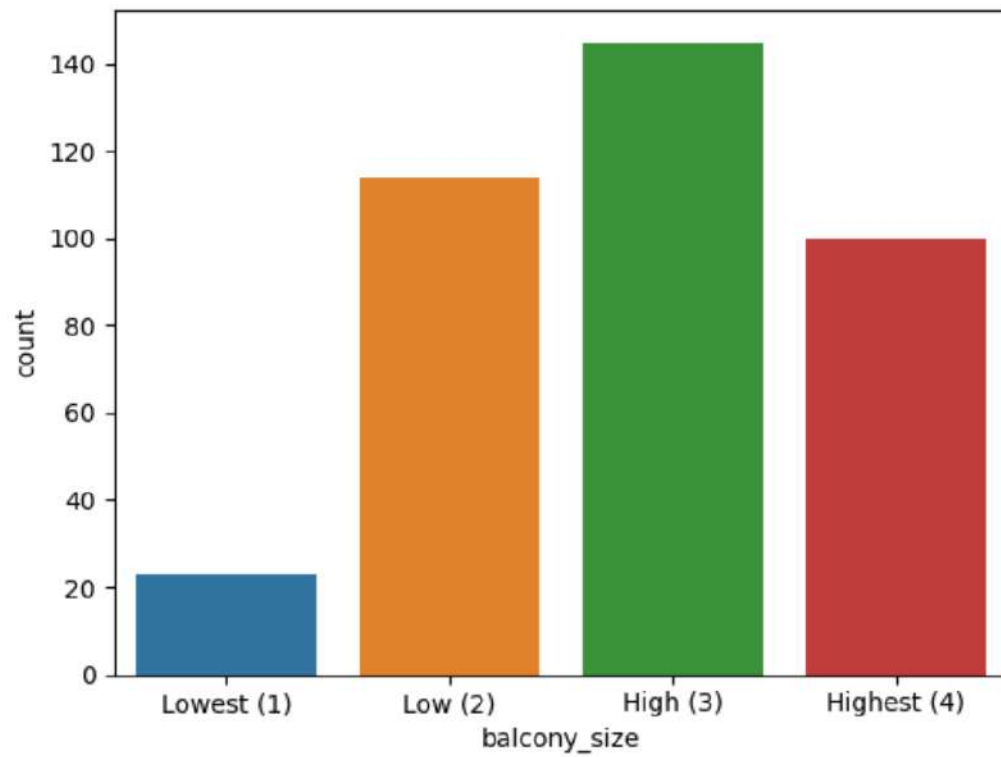


Figure 3.22 Balcony distribution

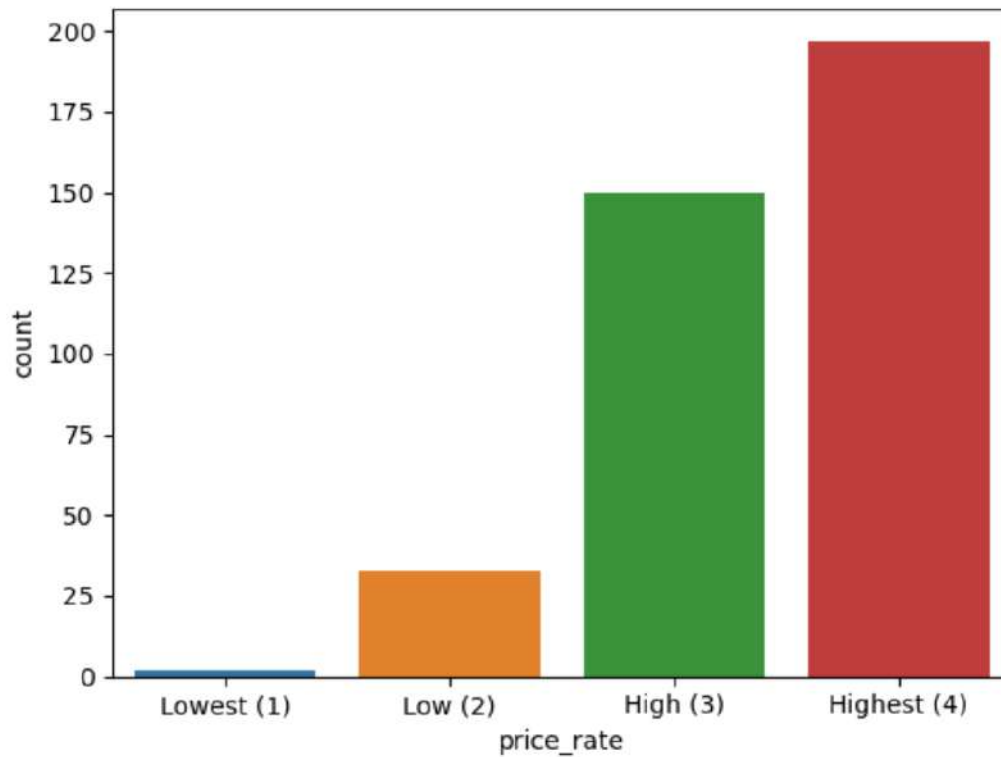


Figure 3.23 Price rate distribution

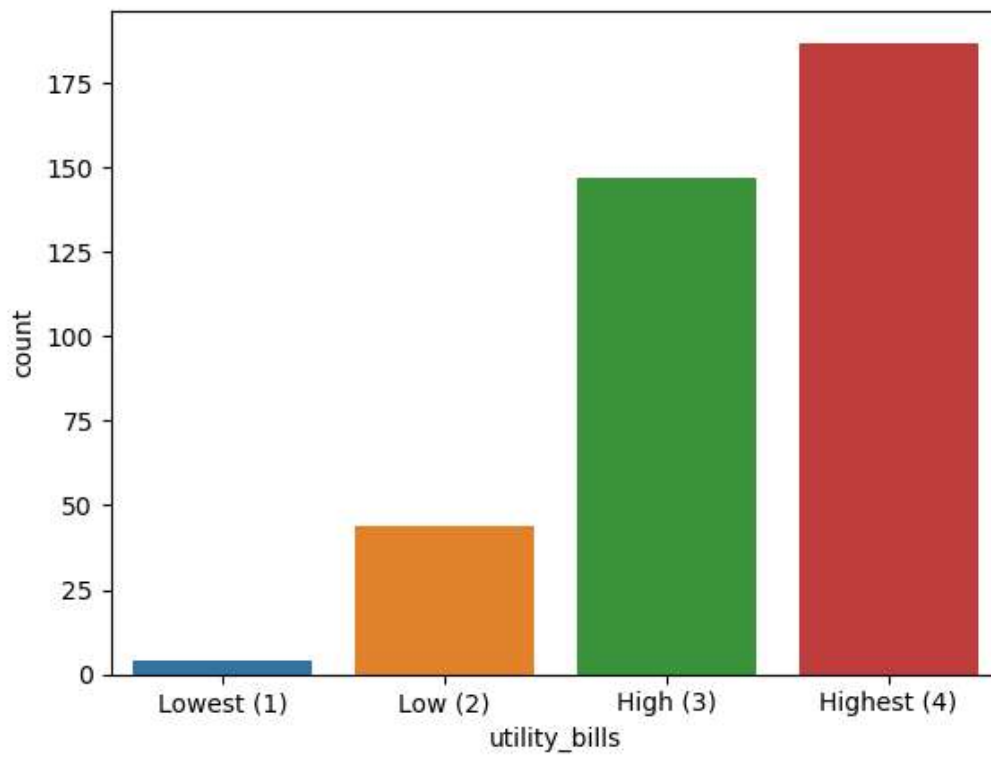


Figure 3.24 Utility bills distribution

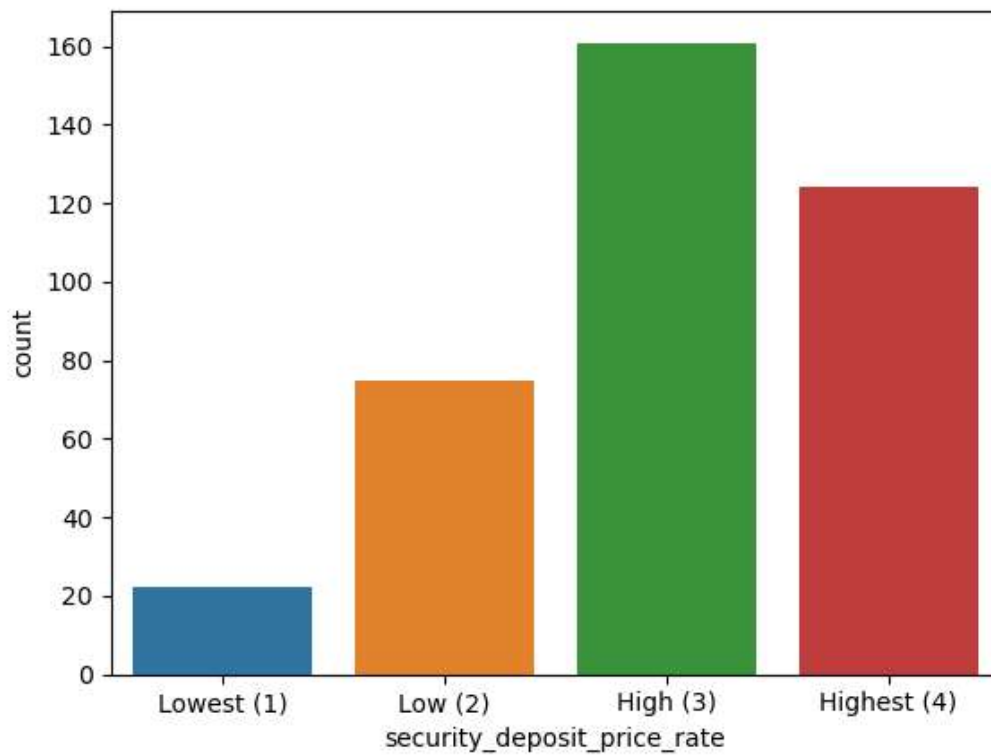


Figure 3.25 Security deposit price distribution

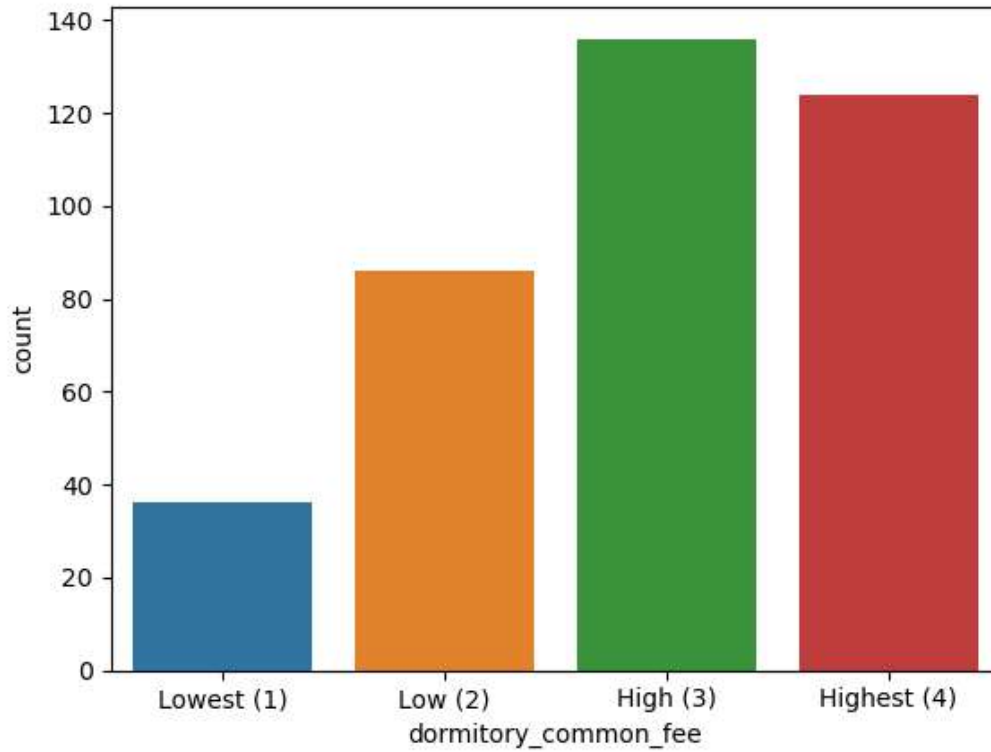


Figure 3.26 Domicile commofee distribution

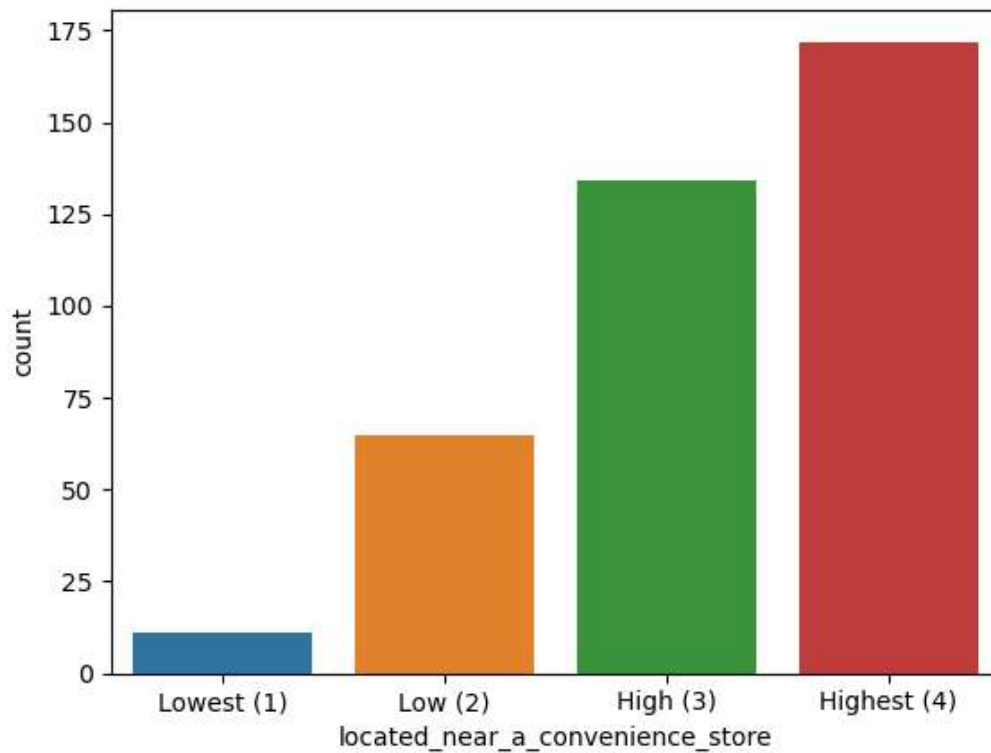


Figure 3.27 Locate near convenience distribution

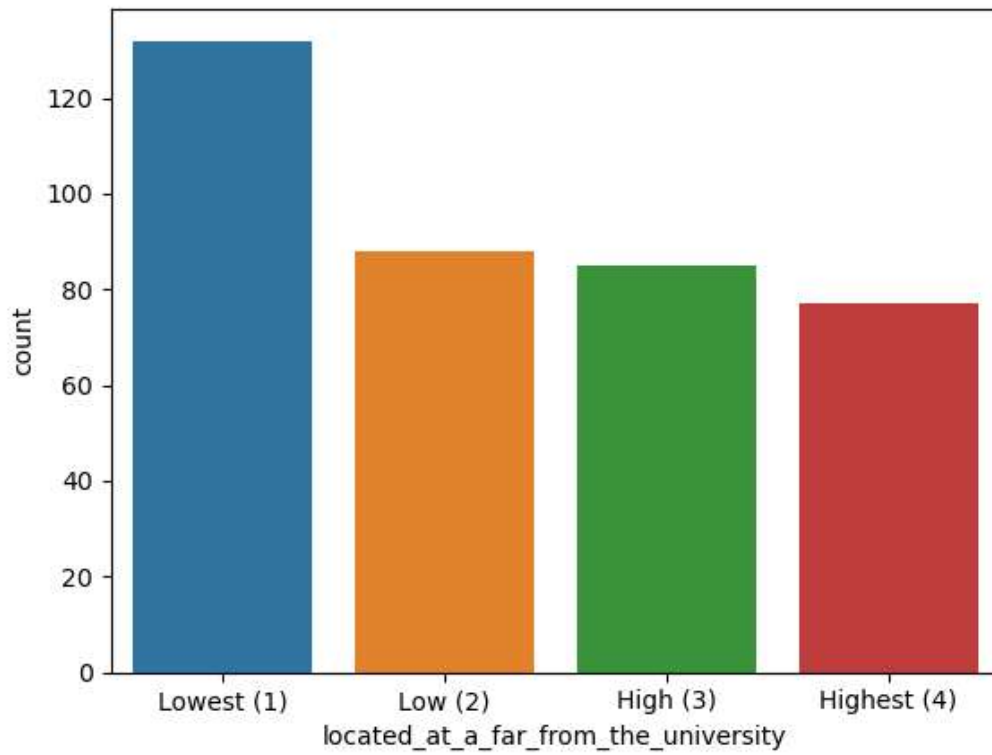


Figure 3.28 Locate far from MFU distribution

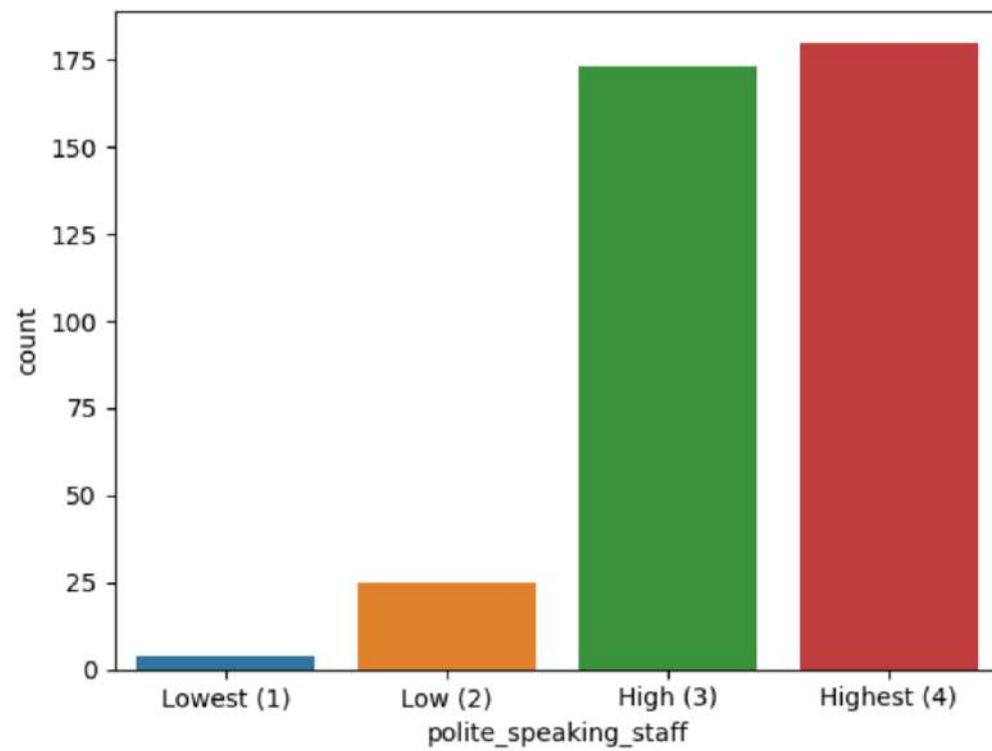


Figure 3.29 Polite speaking staff distribution

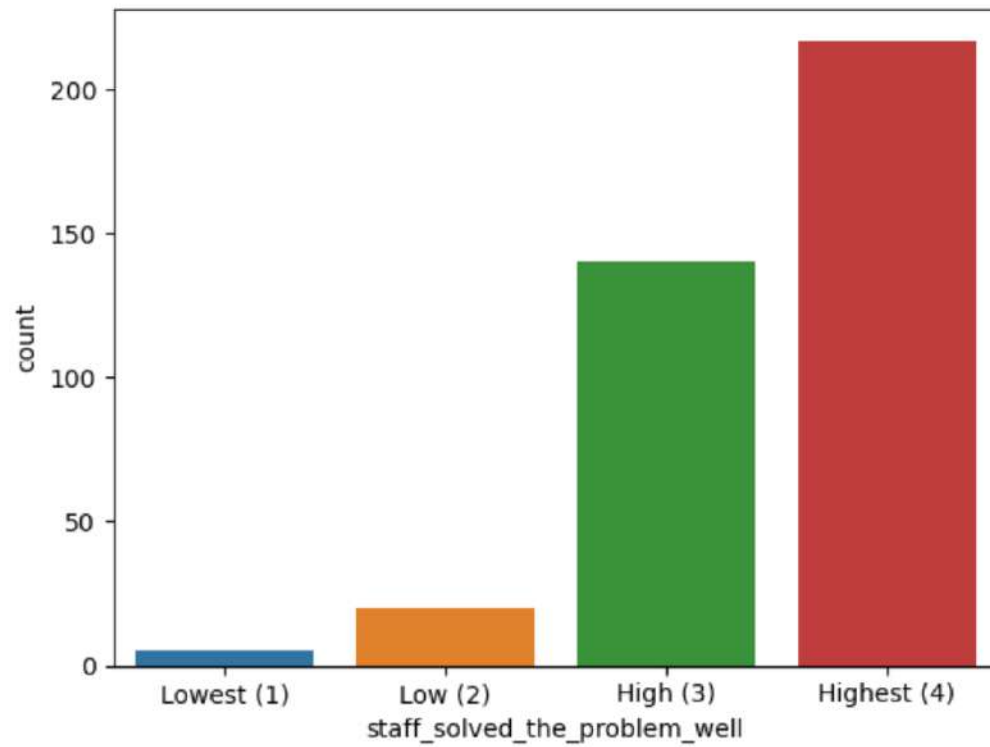


Figure 3.30 Staff solved problem well

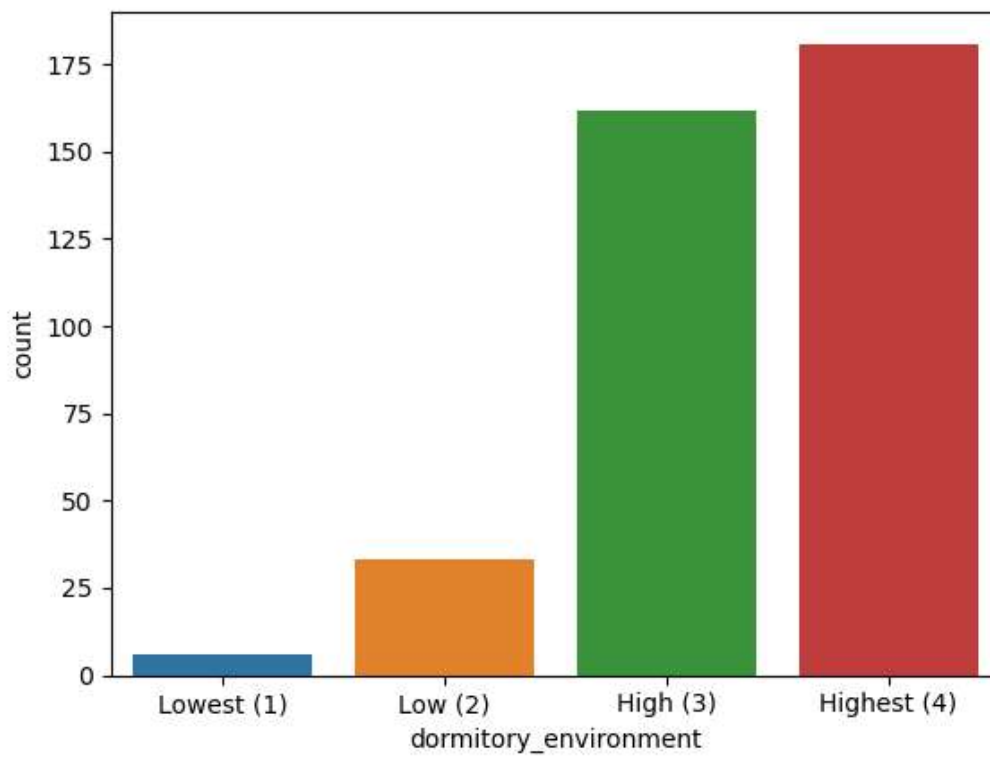


Figure :3.31 Dormitory environment distribution

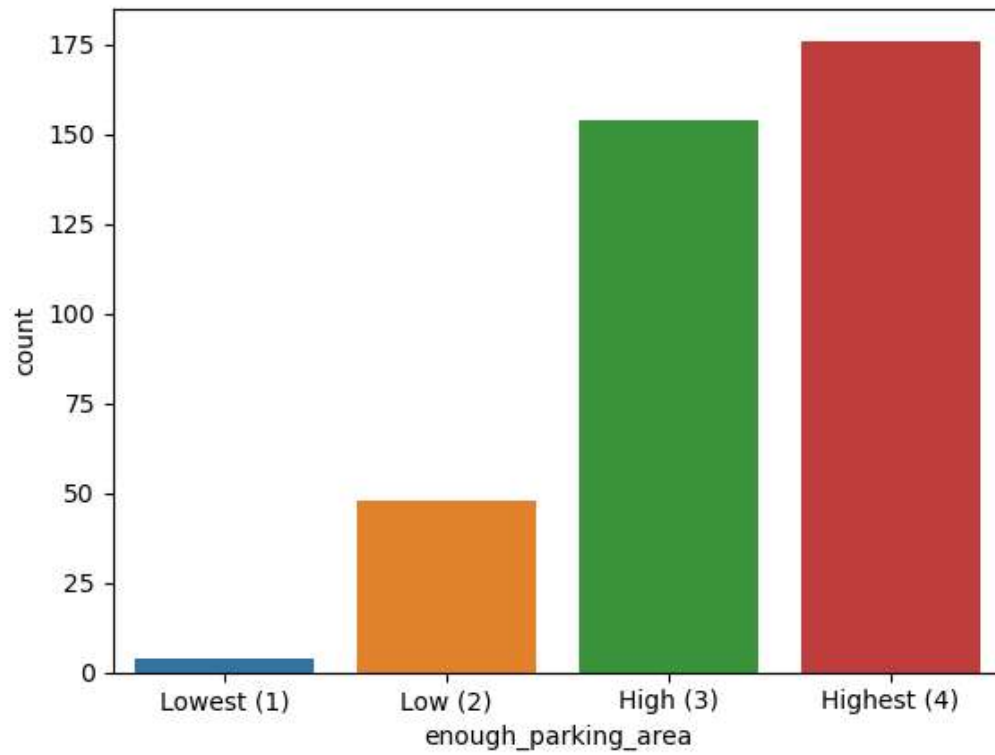


Figure 3.32 Parking area distribution

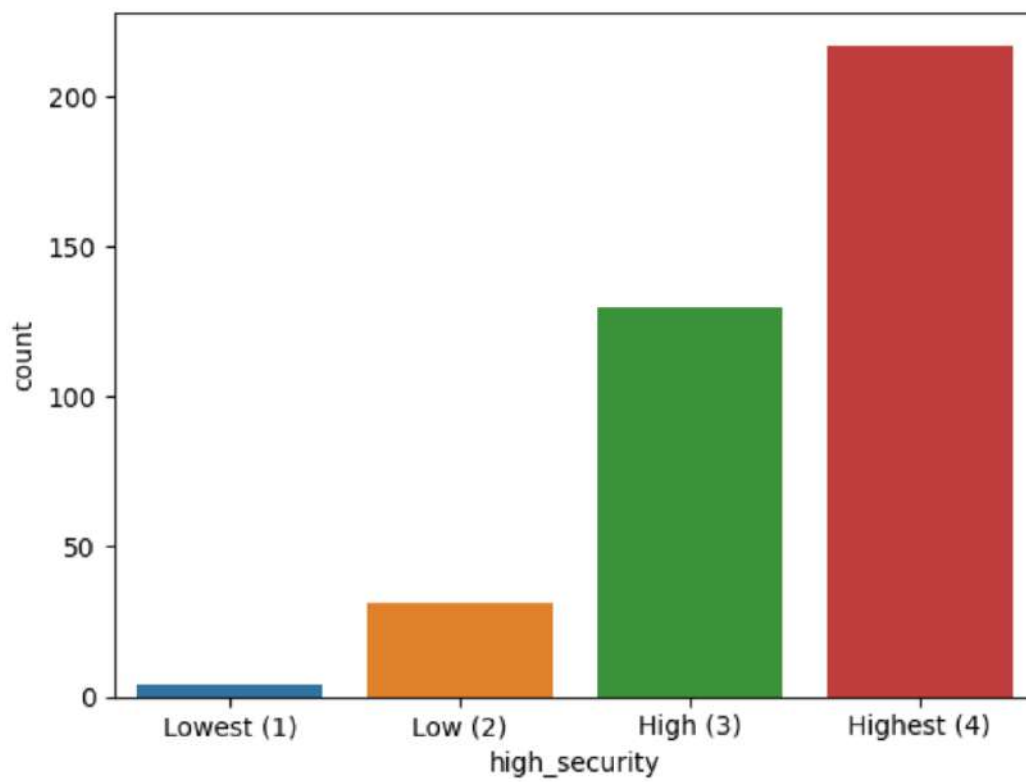


Figure 3.33 High security distribution

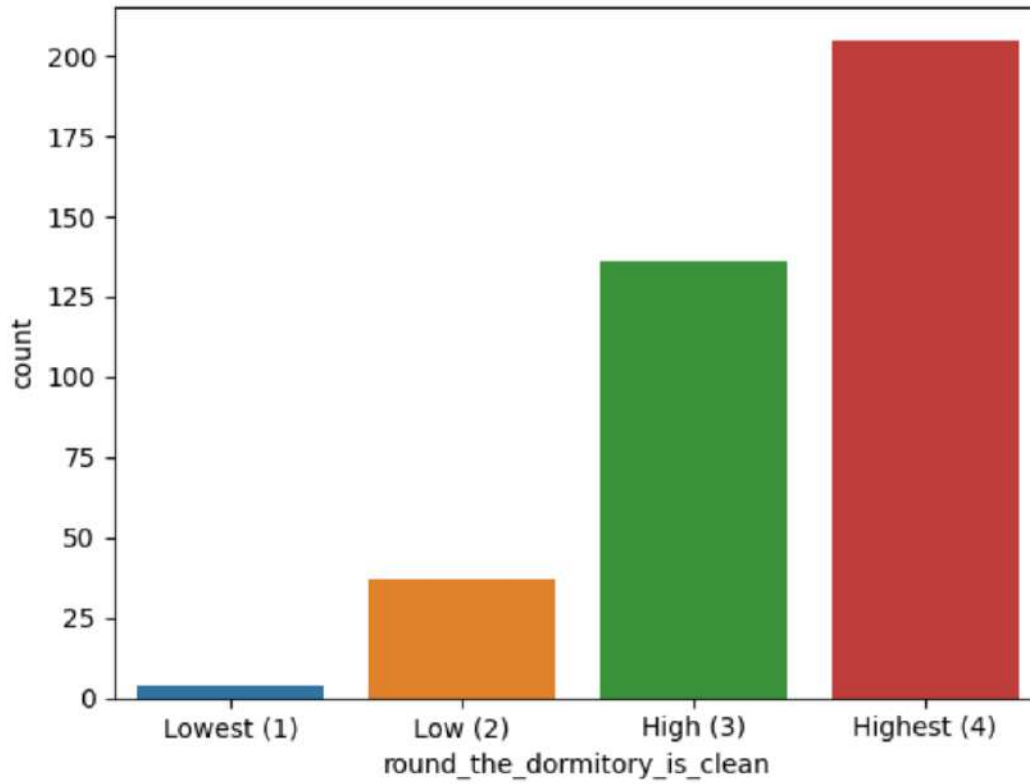


Figure 3.34 Cleanliness around dormitory distribution

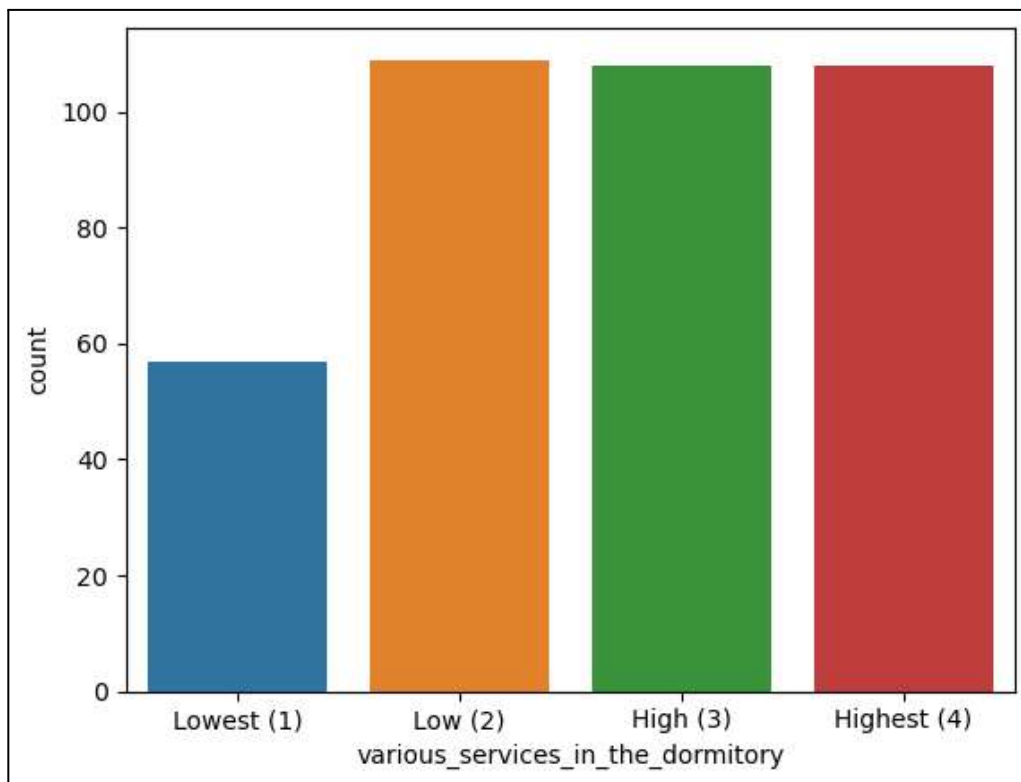


Figure 3.35 Service distribution

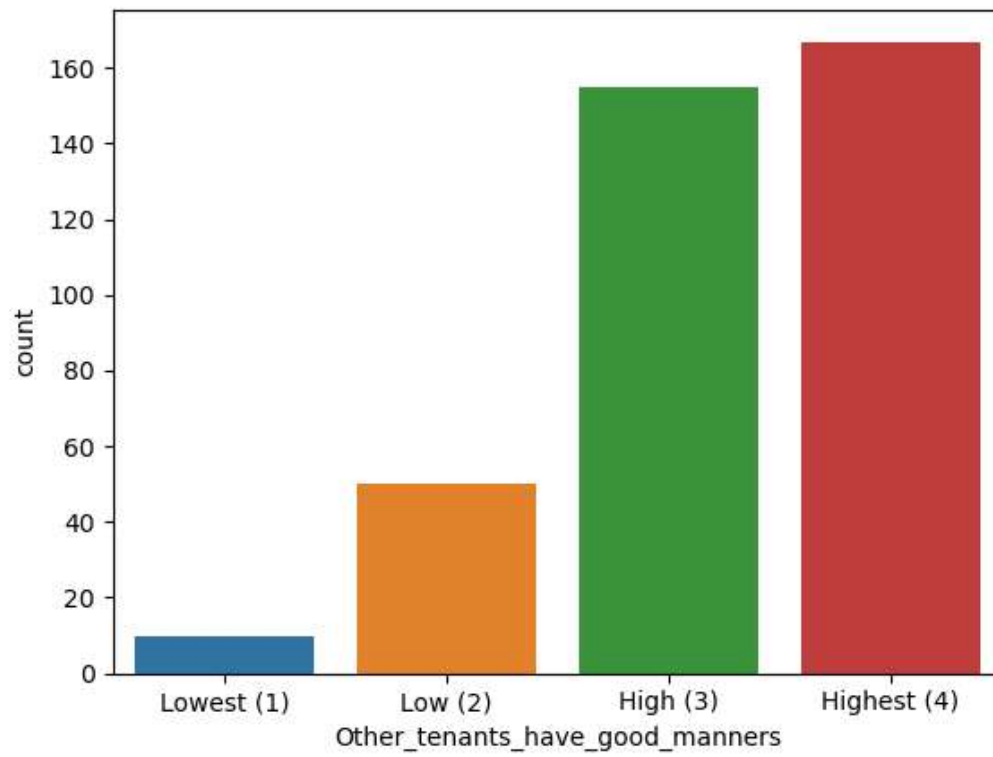


Figure 3.36 Other tenants are good manners distribution

3.5 Machine Learning Process

The machine learning process involves four main steps that consist of data processing, splitting data, training models, and evaluating results. Figure 3.37 shows the Machine Learning process

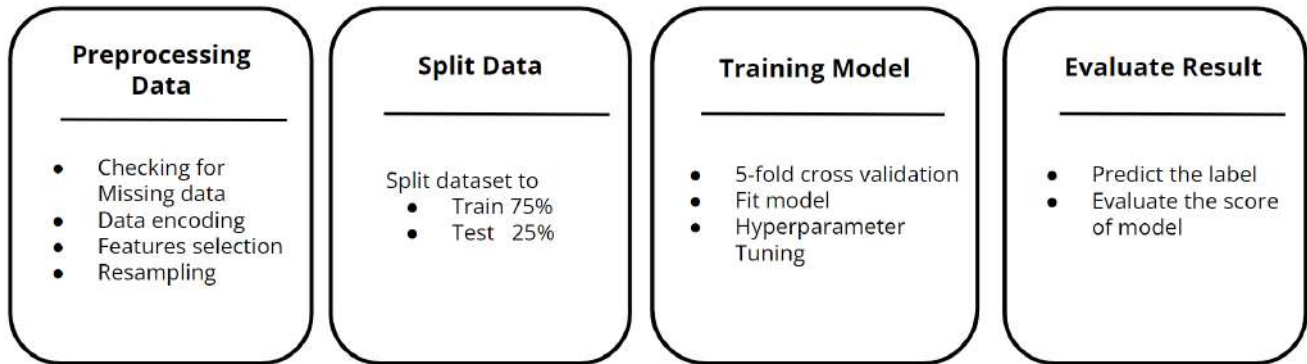


Figure 3.37 Machine Learning Process

3.5.1 Data Preprocessing

Preprocessing is a crucial step in preparing data for machine learning models. This step involves cleaning and transforming the data to ensure that it is in a format suitable for the model to process. In this study, we applied several preprocessing steps to the dataset, including encoding categorical variables feature selection and Resampling data. Although our dataset did not have any missing values, these preprocessing steps are still necessary to improve the accuracy and performance of the machine learning model. In the following sections, we will provide a detailed explanation of each preprocessing step and discuss their impact on the final performance of the model. Figure 3.38 shows the Preprocessing workflow

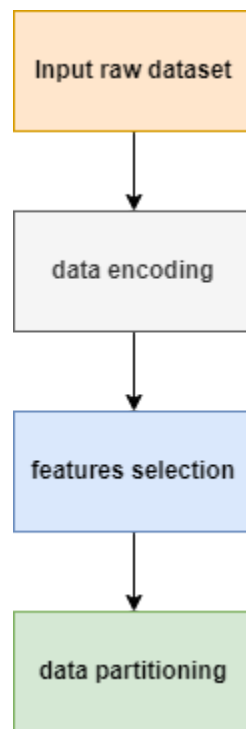


Figure 3.38 Preprocessing flow

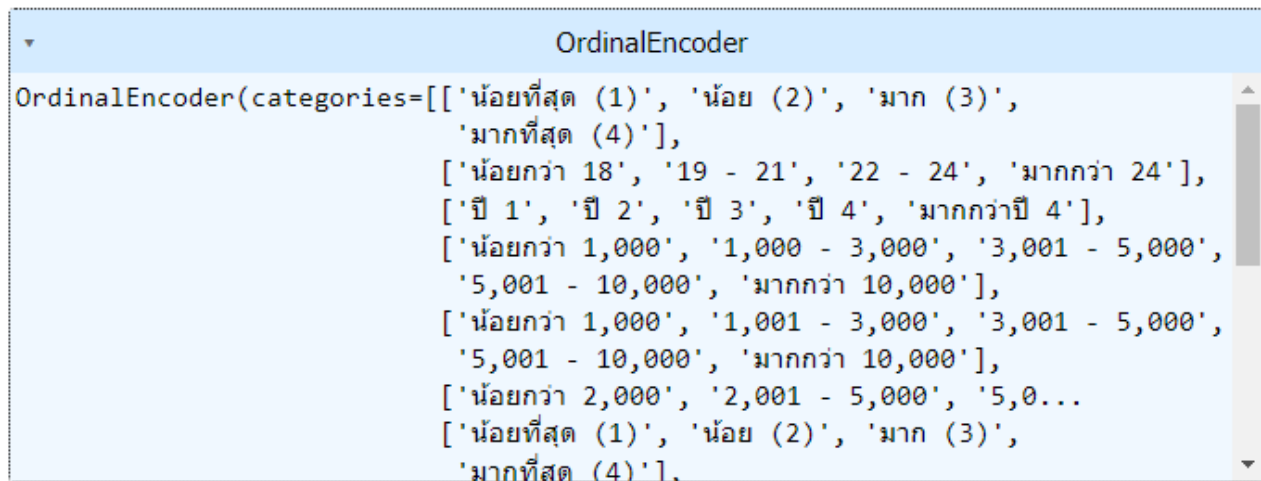
3.5.2 Data Encoding

To prepare the dataset for machine learning, we used data encoding techniques to convert the categorical variables in the dataset to a numerical format. The dataset contained two types of categorical variables: ordinal and nominal.

1.) Ordinal features

For ordinal variables, we utilized the `OrdinalEncoder` class from the `scikit-learn` library [15] to encode categorical values as integers. The `OrdinalEncoder` works by mapping categorical levels to integers in a way that preserves the inherent order of the categories. This encoding method was particularly suitable for variables where the order of the categories held significance, allowing us to represent the data appropriately for models to capture the ordinal relationships within the features. Figure 3.39 shows the `Ordinal Encoder`

```
ordinal_encoder = sklearn.preprocessing.OrdinalEncoder(categories=ordinal_mapping)
ordinal_encoder
```



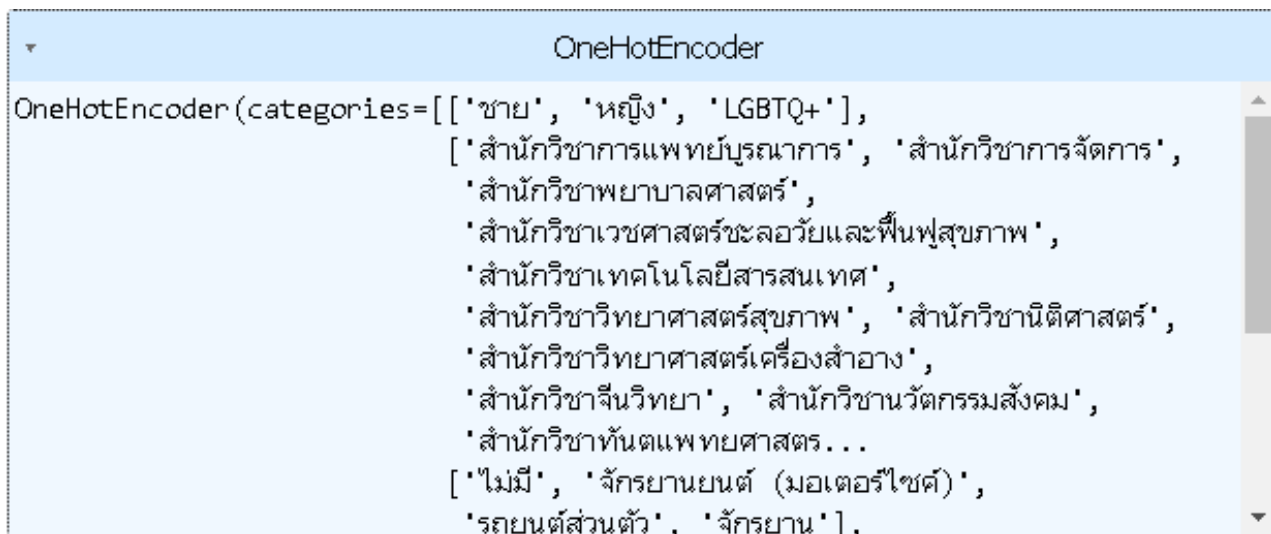
```
OrdinalEncoder(categories=[['น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)',
                             'มากที่สุด (4)'],
                        ['น้อยกว่า 18', '19 - 21', '22 - 24', 'มากกว่า 24'],
                        ['ปี 1', 'ปี 2', 'ปี 3', 'ปี 4', 'มากกว่าปี 4'],
                        ['น้อยกว่า 1,000', '1,000 - 3,000', '3,001 - 5,000',
                         '5,001 - 10,000', 'มากกว่า 10,000'],
                        ['น้อยกว่า 1,000', '1,001 - 3,000', '3,001 - 5,000',
                         '5,001 - 10,000', 'มากกว่า 10,000'],
                        ['น้อยกว่า 2,000', '2,001 - 5,000', '5,001 - 10,000',
                         'มากกว่า 10,000'],
                        ['น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)',
                         'มากที่สุด (4)']])
```

Figure 3.39 Ordinal Encoder

2.) Nominal features

For nominal variables, we employed the OneHotEncoder class from the scikit-learn library [15] to transform categorical values into binary features. The OneHotEncoder operates by creating binary columns for each unique category within the variable, assigning a '1' to the presence of a category and '0' otherwise. This encoding method was well-suited for variables where the order of the categories held no inherent meaning, allowing us to represent categorical data appropriately without imposing any ordinal relationships among the features. Figure 3.40 shows the OneHotEncoder

```
onehot_encoder = sklearn.preprocessing.OneHotEncoder(categories=nominal_mapping, sparse_output=False)
onehot_encoder
```



```
OneHotEncoder(categories=[['ชาย', 'หญิง', 'LGBTQ+'],
                           ['สำนักวิชาการแพทย์บูรณาการ', 'สำนักวิชาการจัดการ',
                            'สำนักวิชาพยาบาลศาสตร์',
                            'สำนักวิชาเวชศาสตร์ชะลอวัยและฟื้นฟูสุขภาพ',
                            'สำนักวิชาเทคโนโลยีสารสนเทศ',
                            'สำนักวิชาวิทยาศาสตร์สุขภาพ', 'สำนักวิชานิติศาสตร์',
                            'สำนักวิชาวิทยาศาสตร์เครื่องสำอาง',
                            'สำนักวิชาจีนวิทยา', 'สำนักวิชานวัตกรรมสังคม',
                            'สำนักวิชาทันตแพทยศาสตร...'],
                           ['ไม่มี', 'จักรยานยนต์ (มอเตอร์ไซด์)',
                            'รถยนต์ส่วนตัว', 'จักรยาน']])
```

Figure 3.40 OneHotEncoder

No	Features Name	Description	Features Class
1	age (Ordinal feature)	age of student	น้อยกว่า 18, 19 - 21, 22 - 24, มากกว่า 24
2	education_level (Ordinal feature)	education level of student	ปี 1, ปี 2, ปี 3, ปี 4, มากกว่าปี 4
3	average_monthly_income (Ordinal feature)	average income that student got in each month	น้อยกว่า 1,000 , 1,000 - 3,000, 3,001 - 5,000, 5,001 - 10,000, มากกว่า 10,000
4	average_monthly_expenditure (Ordinal feature)	average expend that student got in each month	น้อยกว่า 1,000 , 1,000 - 3,000, 3,001 - 5,000, 5,001 - 10,000, มากกว่า 10,000
5	security_deposit_price (Ordinal feature)	security deposit price of accommodation	น้อยกว่า 2,000, 2,001 - 5,000 , 5,001 - 8,000 , มากกว่า 8,001
6	facilities (Ordinal feature)	facilities of accommodation	ห้องเปล่าไม่มีเฟอร์นิเจอร์ (Empty room)', 'ห้องมีสิ่งอำนวยความสะดวกบางส่วน (Semi-Furnished) (ควรมีแอร์, ทีวี, โต๊ะเครื่องแป้ง, เตียง, ตู้เย็น, เครื่องทำน้ำอุ่น, ระเบียงห้อง เป็นต้น)', 'ห้องมีสิ่งอำนวยความสะดวกครบ (Fully-Furnished) (ควรมีโซฟาห้องครัว, ห้องนั่งเล่น, Wi-Fiส่วนตัว เป็นต้น)'
7	travel_time (Ordinal feature)	time that student use to go to university	น้อยกว่า 5 นาที', '6 - 10 นาที', '10 - 15 นาที', '16 - 20 นาที', 'มากกว่า 20 นาที'
8	distance_to_university (Ordinal feature)	distance from accommodation to university	น้อยกว่า 1 กิโลเมตร', '1 - 2 กิโลเมตร', '2 - 3 กิโลเมตร', '2 - 4 กิโลเมตร', '4 - 5 กิโลเมตร', 'มากกว่า 5 กิโลเมตร'
9	room_size (Ordinal feature)	room size of accommodation	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
10	cleanliness (Ordinal feature)	how clean of accommodation	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
11	amenities (Ordinal feature)	amenities in	น้อยที่สุด (1)', 'น้อย (2)', 'มาก

		accommodation	(3)', 'มากที่สุด (4)'
12	decoration (Ordinal feature)	decoration in accommodation	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
13	balcony_size (Ordinal feature)	balcony size in accommodation	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
14	price_rate (Ordinal feature)	price rate of accommodation	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
15	utility_bills (Ordinal feature)	utility bills from accommodation	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
16	security_deposit_price_rate (Ordinal feature)	security deposit price rate of accommodation	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
17	dormitory_common_fee (Ordinal feature)	dormitory common fee	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
18	located_near_a_convenience_store (Ordinal feature)	how close of accommodation and store	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
19	located_at_a_far_from_the_university (Ordinal feature)	how far of accommodation and university	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
20	polite_speaking_staff (Ordinal feature)	staff have a polite conversation	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
21	staff_solved_the_problem_well (Ordinal feature)	staff can solved problem as well	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
22	dormitory_environment (Ordinal feature)	environment of accommodation	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
23	enough_parking_area (Ordinal feature)	parking of accommodation enough for student	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
24	high_security (Ordinal feature)	accommodation have a high security	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
25	round_the_dormitory_is_clean (Ordinal feature)	around of accommodation area is clean	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
26	various_services_in_the_dormitory (Ordinal feature)	service of accommodation	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
27	Other_tenants_have_good_manners (Ordinal feature)	other living have a good manner	น้อยที่สุด (1)', 'น้อย (2)', 'มาก (3)', 'มากที่สุด (4)'
28	sex (Nominal feature)	sex of student	Men, Women, LGBTQ+

29	school (Nominal feature)	like a major that student study	สำนักวิชาการแพทย์บูรณาการ, สำนักวิชาการจัดการ, สำนักวิชาพยาบาลศาสตร์, สำนักวิชาเวชศาสตร์ชะลอวัยและฟื้นฟูสุขภาพ, สำนักวิชาเทคโนโลยีสารสนเทศ, สำนักวิชาวิทยาศาสตร์สุขภาพ, สำนักวิชานิติศาสตร์, สำนักวิชาวิทยาศาสตร์เครื่องสำอาง, สำนักวิชาจีนวิทยา, สำนักวิชานวัตกรรมสังคม, สำนักวิชาทันตแพทยศาสตร์, สำนักวิชาศิลปศาสตร์, สำนักวิชาอุตสาหกรรมเกษตร, สำนักวิชาวิทยาศาสตร์, สำนักวิชาแพทยศาสตร์
30	vehicle_possession (Nominal feature)	vehicle that student have	ไม่มี, จักรยานยนต์ (มอเตอร์ไซค์), รถยนต์ส่วนตัว, จักรยาน
31	travel_to_university_method (Nominal feature)	the way that student use to go to university	จักรยานยนต์ (มอเตอร์ไซค์), รถยนต์ส่วนตัว, รถเขียว, เดิน, จักรยาน
32	domicile (Nominal feature)	domicile of student	ภาคเหนือ, ภาคกลาง, ภาคตะวันออก, ภาคใต้
33	dormitory_location (Nominal feature)	accommodation location	'ใกล้มหาวิทยาลัย', 'ใกล้รถสาธารณะ', 'ใกล้ร้านอาหาร', 'ใกล้เซเว่น', 'ใกล้สถานที่ท่องเที่ยว', 'อื่นๆ'
34	gained_info_by (Nominal feature)	gained info of accommodation by	'ติดต่อด้วยตัวเอง', 'เพื่อน', 'โซเชียลมีเดีย', 'เจ้าของหอพัก', 'website', 'อื่นๆ'

Table 3.3 Data encoder types

3.2.3 Sample selections

Addressing class imbalance within the dataset is a critical aspect of enhancing the performance and reliability of machine learning models. The inherent imbalance, where certain classes are significantly underrepresented compared to others, poses challenges in effectively training models. In pursuit of improved model performance, two distinct methodologies were employed to mitigate the effects of class imbalance.

3.2.3.1 Exclusion of Class 3 due to Limited Sample Size and Outliers

The initial step in handling the imbalance involved an in-depth examination of the dataset's classes. Class 3 emerged with a limited number of samples, totaling only 13 records, and exhibited a notable presence of outliers. This subsection delves into the rationale behind excluding Class 3 from the analysis due to its sparse representation and the potential influence of outliers on model learning. Figure 3.41 shows the process of Exclusion on Class 3.

```
data_train_test = (
    data_train_test_ordinal_features
        .join(data_train_test_nominal_features)
        .join(data_train_test_label)
)
data_train_test = data_train_test[data_train_test['rental_price'] != 3.0]
data_train_test['rental_price'].value_counts()
```

```
1.0    258
2.0     88
0.0     23
Name: rental_price, dtype: int64
```

Figure 3.41 : Exclusion of class 3 process

3.2.4 Feature Selection

To select the most important features for our machine learning model, we used `SequentialFeatureSelector` from the `scikit-learn` library[15]. This algorithm sequentially adds and removes features based on their performance in a given metric, and we used the Logistic regression scoring function to evaluate feature importance. Figure 3.42 shows the Sequential features selection process

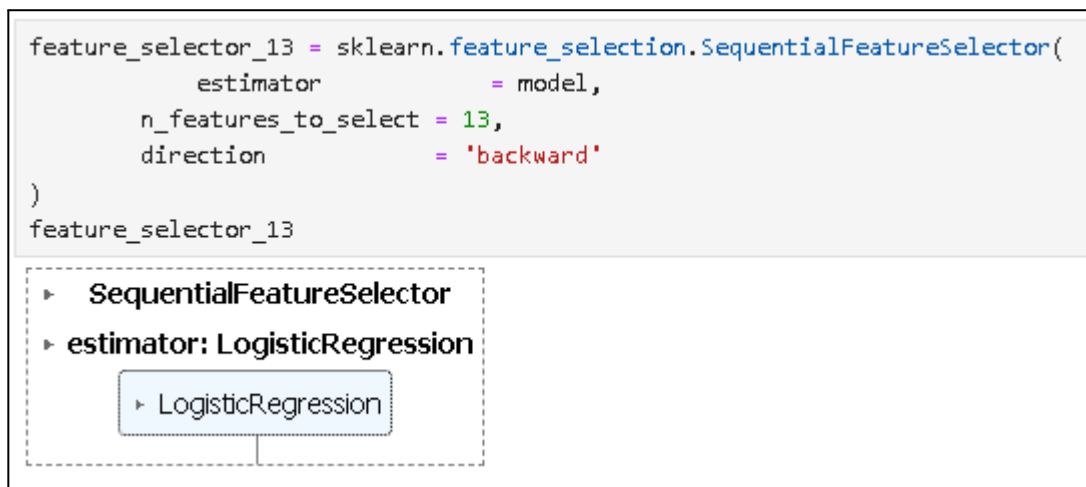


Figure 3.42 Feature Selection

The number of features selected by SequentialFeatureSelector was optimized using cross-validation to ensure that the model did not overfit the training data. We used a grid search to evaluate the expected number of features to select, and found that 13 features were expected for our model as can be seen in table 3.4

No	Feature Name
1	education_level
2	security_deposit_price
3	facilities
4	distance_to_university
5	decoration
6	balcony_size
7	polite_speaking_staff
8	round_the_dormitory_is_clean
9	school_สำนักวิชาวิทยาศาสตร์เครื่องสำอาง
10	school_สำนักวิชานวัตกรรมสังคม
11	domicile_ภาคเหนือ
12	gained_info_by_เพื่อน
13	gained_info_by_เจ้าของหอพัก

Table 3.4 Selected 13 Features

The final model was trained on the selected features, which were determined to have the highest impact on predicting the accommodation price range.

3.2.5 Splitting data

To prepare the data for model training, the dataset was partitioned into training and testing sets using the `train_test_split` function from `scikit-learn`[15]. This method splits the dataset into input features (X) and the target variable (y).

To ensure an equitable distribution of classes in both sets, the parameter `stratify=y` was applied, maintaining the proportion of classes from the original dataset. Approximately 30% of the data was assigned to the testing set (X_test, y_test), while the remaining 70% constituted the training set (X_train, y_train). Additionally, to prevent any bias from the data order, the `shuffle=True` parameter randomized the samples before the split. To maintain consistency in the split across different runs, a `random_state` value of 42 was set, ensuring reproducibility in subsequent executions of the code. Figure 3.43 shows the splitting data parameters

```
# Split the data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    stratify=y,
    test_size=0.3,
    shuffle=True,
    random_state=42)
```

Figure 3.43 Splitting data

3.2.6 Model Training

After preprocessing the dataset through data encoding and feature selection then splitting method, After that we trained a machine learning model to predict the accommodation price range. The training process involved three main methods: Input train data, hyperparameter tuning and model creation. Figure 3.44 shows the model training workflow

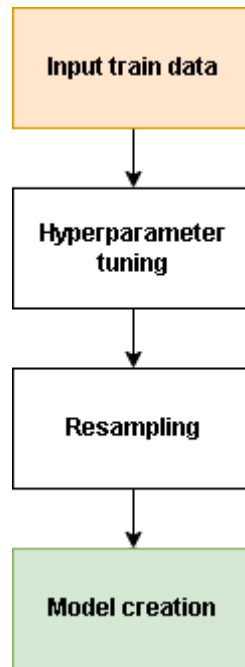


Figure 3.44 Model Training workflow

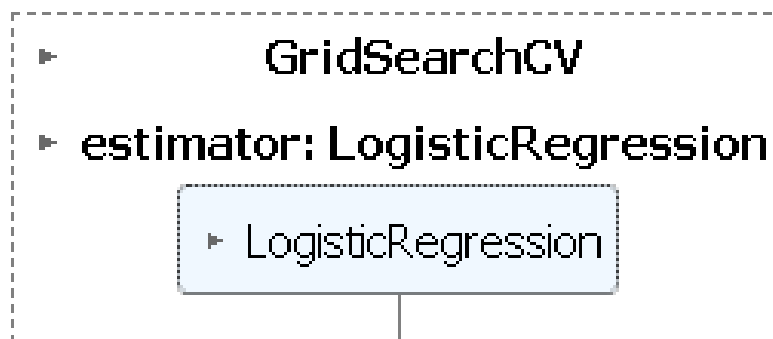
3.2.7 Hypermeter tuning

To optimize our model's hyperparameters, we employed GridSearchCV from the scikit-learn library, conducting an exhaustive search across a specified parameter space. Implementing 5-fold cross-validation within GridSearchCV allowed systematic evaluation of hyperparameter combinations, aiding in the identification of the most effective settings that maximized model performance. This approach not only fine-tuned the model but also ensured its robustness by rigorously assessing performance across five distinct validation folds.

The parameter 'penalty' was set to 'l2' during the grid search process. The choice of 'l2' over 'l1' was informed by its ability to offer more robustness and better predictive accuracy at a nominal expense for additional computational resources. The remaining parameters were determined based on their support for 'penalty=l2', ensuring a comprehensive exploration within the chosen penalty space. Figure 3.45 shows the Hyperparameter Tuning process

```
clf = LogisticRegression( max_iter = 1000)
param_grid = {
    'penalty': ['l2'],
    'solver': [ 'newton-cg'],
    'C': [0.1,1,10],
    'multi_class': ['ovr','multinomial'],
    'fit_intercept' : [True,False],
}

grid_search = GridSearchCV(clf, param_grid=param_grid, cv=5, verbose=True, n_jobs=-1)
grid_search.fit(X_train_resampled, y_train_resampled)
```



```
print("Best parameters: {}".format(grid_search.best_params_))
```

```
Best parameters: {'C': 10, 'fit_intercept': False, 'multi_class': 'multinomial', 'penalty': 'l2', 'solver': 'newton-cg'}
```

Figure 3.45 Hyperparameter Tuning

3.2.8 Imbalanced data

We addressed the imbalanced distribution of the target variable by using the undersampling technique called NearMiss on class 1 (2,001-5,000). This method is to select a subset of samples from the majority class that are close in distance to samples in the minority class. This helps in reducing the imbalance between the classes while preserving the information present in the dataset. Then we use the oversampling technique called SMOTE in class 0 (Lower than 2,000) and class 2 (5,001-8,000). This method is an oversampling technique used to address class imbalance in machine learning datasets. It focuses on the minority class by generating synthetic samples to balance the class distribution. And we use `sampling_strategy` to fix the amount of dataset records in each class by : class 0 = 62 records, class 1 = 124 records and class 2 = 124 records. Figure 3.46 shows the amount of data records before and after applying resampling technique

Before applying resampling technique

```
1.0    180
2.0     62
0.0     16
Name: rental_price, dtype: int64
```

After applying resampling technique

```
2.0    124
1.0    124
0.0     62
Name: rental_price, dtype: int64
```

Figure 3.46 Imbalanced Data

3.2.8 Model Creation

We developed a machine learning model utilizing the scikit-learn library [15], employing a blend of multiclass classification algorithms including Random Forest, Decision Tree, and Logistic Regression. Each of these algorithms functions to model the probability of a multiclass response variable. The selection of these models was based on their collective capacity to effectively manage both nominal and ordinal encoded features, coupled with their interpretability. To optimize the model's performance, we utilized hyperparameter tuning methods to fine-tune the parameters. Figure 3.47 to figure 3.49 shows the model creation process in each algorithm.

```
model = sklearn.linear_model.LogisticRegression(
    class_weight='balanced',
    penalty      = 'l2',
    fit_intercept = True,
    random_state = 0,
    C = 10,
    multi_class = 'multinomial',
    solver      = 'newton-cg',
)
model
```

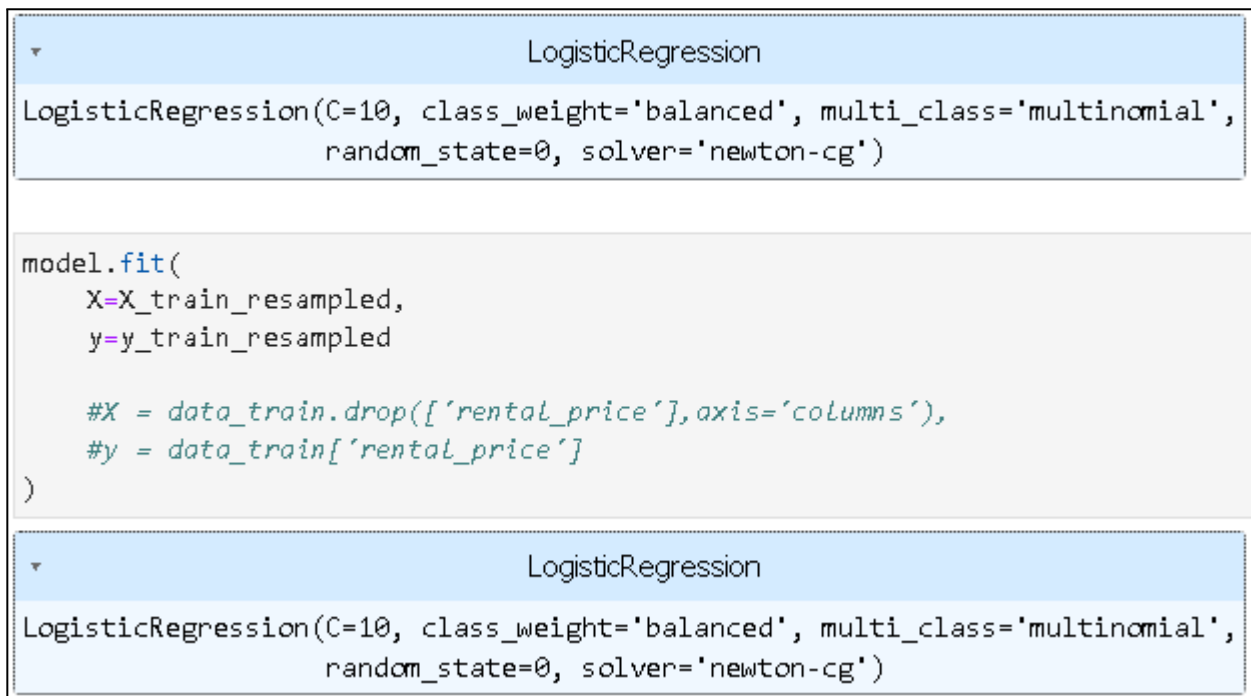


Figure 3.47 Logistic regression model Creation

```
model = sklearn.tree.DecisionTreeClassifier(
    criterion='entropy',
    max_depth=5,
    random_state=0,
)
```

▼ DecisionTreeClassifier
DecisionTreeClassifier(criterion='entropy', max_depth=5, random_state=0)
<pre>model.fit(X=X_train_resampled, y=y_train_resampled)</pre>
▼ DecisionTreeClassifier
DecisionTreeClassifier(criterion='entropy', max_depth=5, random_state=0)

Figure 3.48 Decision tree model Creation

```
model = RandomForestClassifier(
    criterion = 'entropy',
    max_depth = None,
    max_leaf_nodes = None,
    min_samples_leaf = 1,
    min_samples_split = 2,
    n_estimators = 50,
    random_state = None,
)
model
```

▼ RandomForestClassifier
RandomForestClassifier(criterion='entropy', n_estimators=50)
<pre>model.fit(X=X_train_resampled, y=y_train_resampled)</pre>
▼ RandomForestClassifier
RandomForestClassifier(criterion='entropy', n_estimators=50)

Figure 3.49 Random forest model Creation

3.2.9 Model Testing

After model creation, it is essential to evaluate the model's performance before utilizing it in subsequent steps. This evaluation allows us to assess the model's predictive capabilities and make informed decisions. By making predictions using the model on a test dataset, we obtain evaluation results that provide insights into its performance. Through thorough analysis of these results, we can determine the model that demonstrates the best performance, ensuring its suitability for the task at hand. Figure 3.50 shows the process of testing model testing

```
def evaluate():
    for i in ['13','15','18','20','23','25','28','30','35','38','40']:
        data_path = f'../data/split/data_features_selected_{i}.test.csv'
        model_path = f'model/model_6_11_23/model_logistic_selected_{i}_cross_validation.joblib'

        data_test = pandas.read_csv(data_path).set_index('id')
        model = joblib.load(model_path)
        data_test['prediction'] = model.predict(data_test.drop(['rental_price'], axis='columns'))

        print(i+'features'+'\n',sklearn.metrics.classification_report(
            y_true = data_test['rental_price'],
            y_pred = data_test['prediction']
        ))
```

Figure 3.50 Model testing

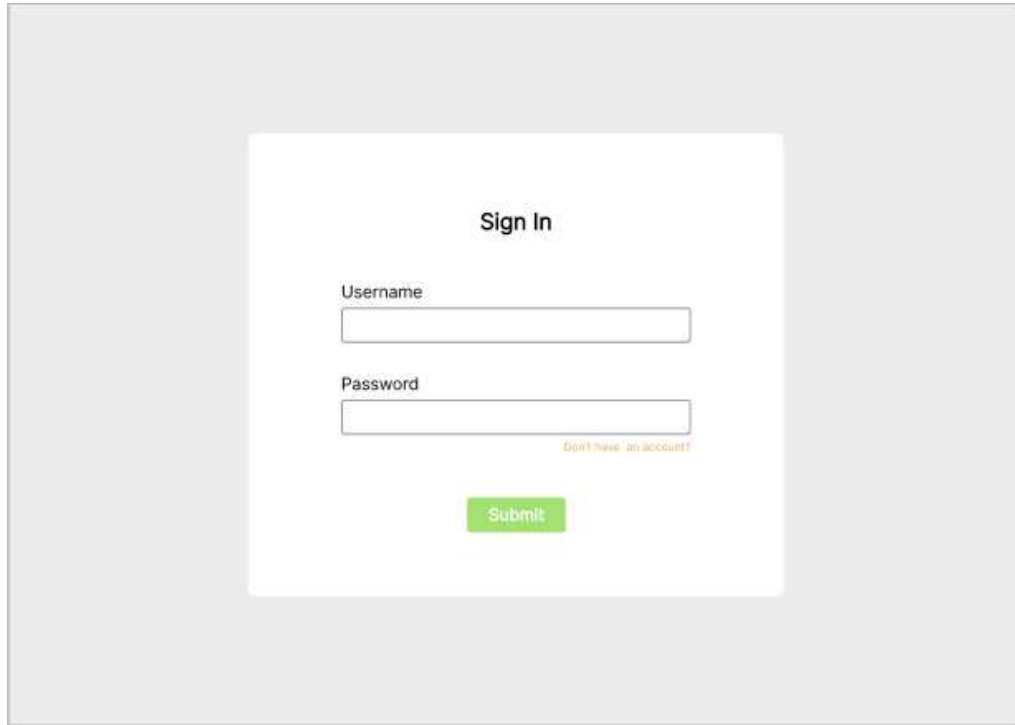
3.2.10 Evaluate Result

After model testing, we utilized the 'classification_report' function from the 'sklearn.metrics' module [15] to assess the predictive performance of our models. Given the inherent class imbalance in our dataset, focusing on metrics such as F1-score and precision could yield biased results. Hence, we emphasized the average recall score provided by the 'classification_report' function. The average recall score offers a more robust evaluation measure for imbalanced datasets, as it considers the recall for each class and provides an aggregated average value, delivering a clearer understanding of our models' ability to capture instances across all classes, despite their imbalanced nature."

3.3 Web application process

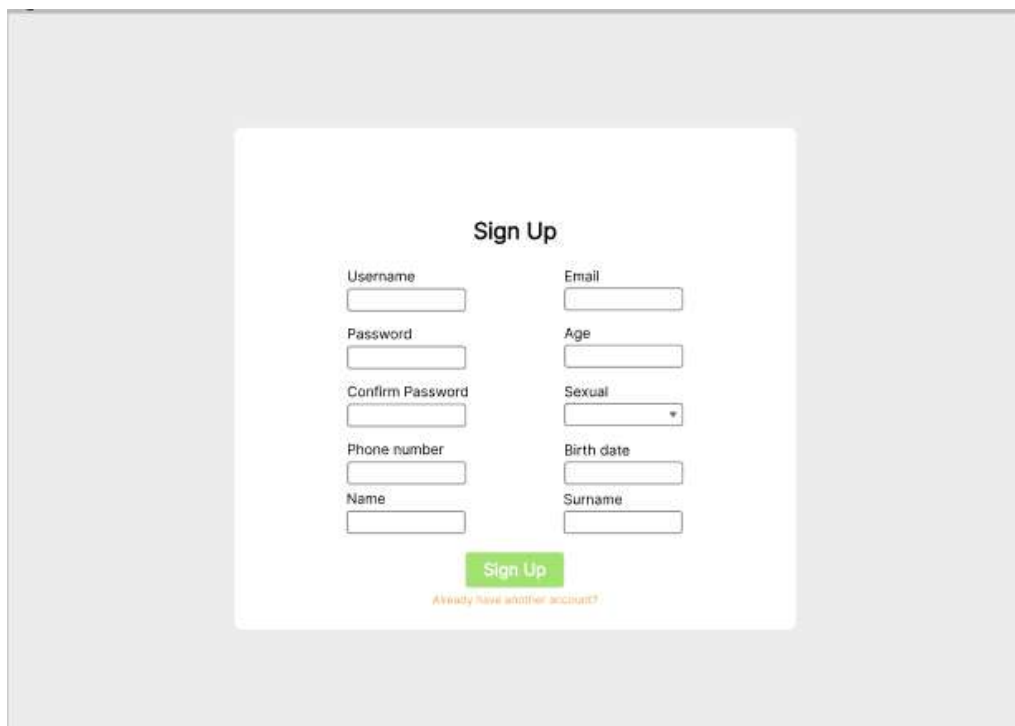
3.3.1 Web application Design

The web application design is illustrated in the following figures: Figure 3.51 to Figure 3.57. These figures represent the web application design of user interface and admin interface.



A login form titled "Sign In" centered on a light gray background. The form is white and contains two input fields: "Username" and "Password". Below the "Password" field is a link that says "Don't have an account?". At the bottom of the form is a green "Submit" button.

Figure 3.51 User interface login design



A registration form titled "Sign Up" centered on a light gray background. The form is white and contains two columns of input fields. The left column includes "Username", "Password", "Confirm Password", "Phone number", and "Name". The right column includes "Email", "Age", "Sexual" (a dropdown menu), "Birth date", and "Surname". At the bottom of the form is a green "Sign Up" button. Below the button is a link that says "Already have another account?".

Figure 3.52 User interface register design

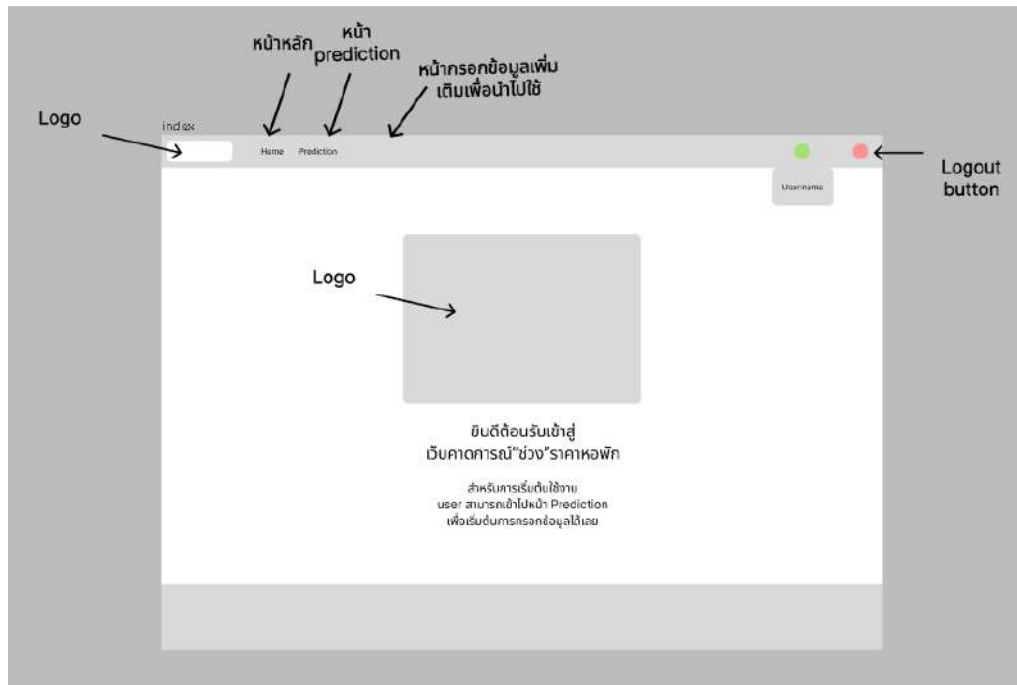


Figure 3.53 User interface home design

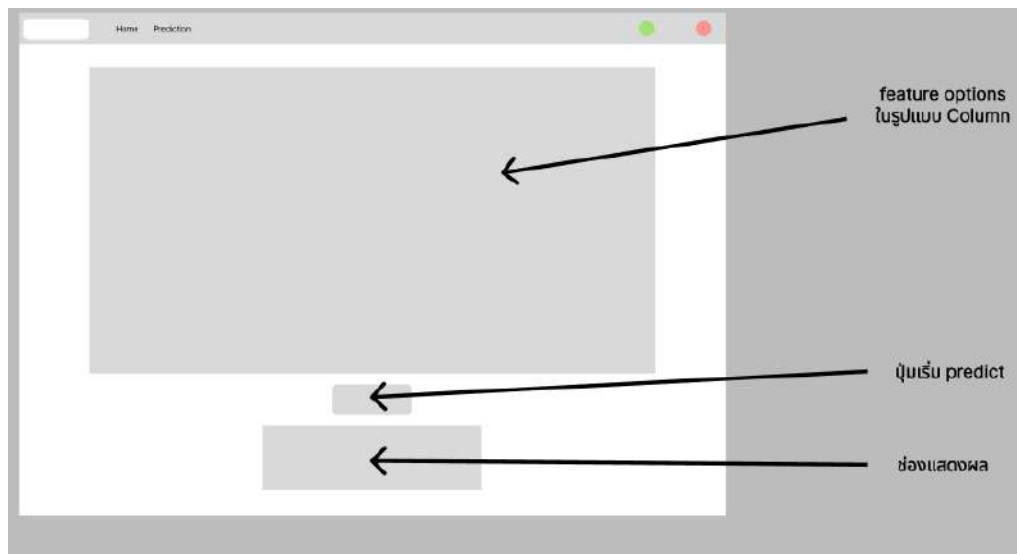


Figure 3.54 User interface prediction design

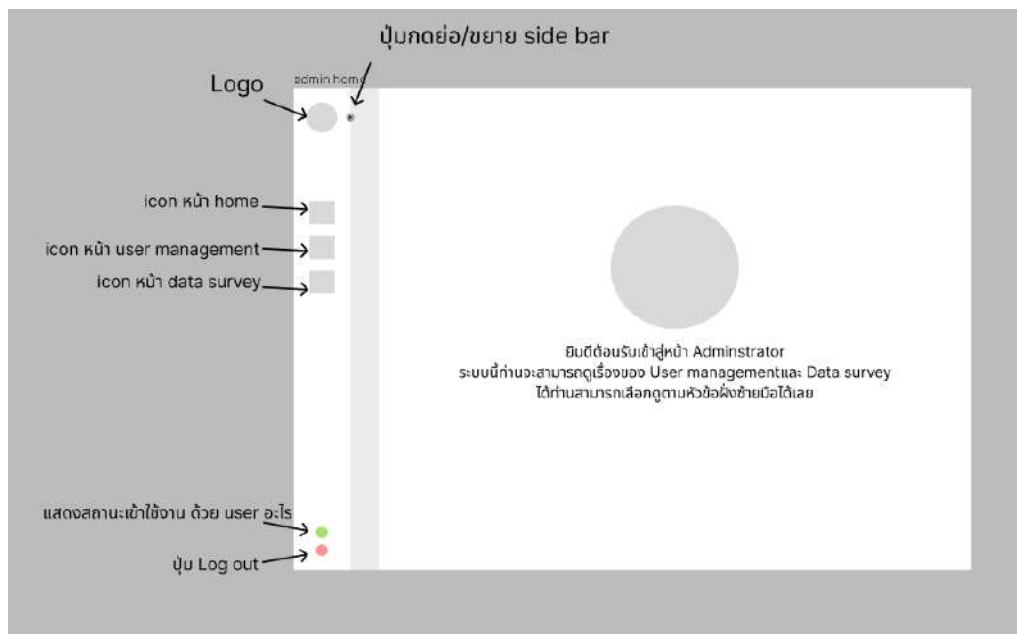


Figure 3.55 Admin interface home design

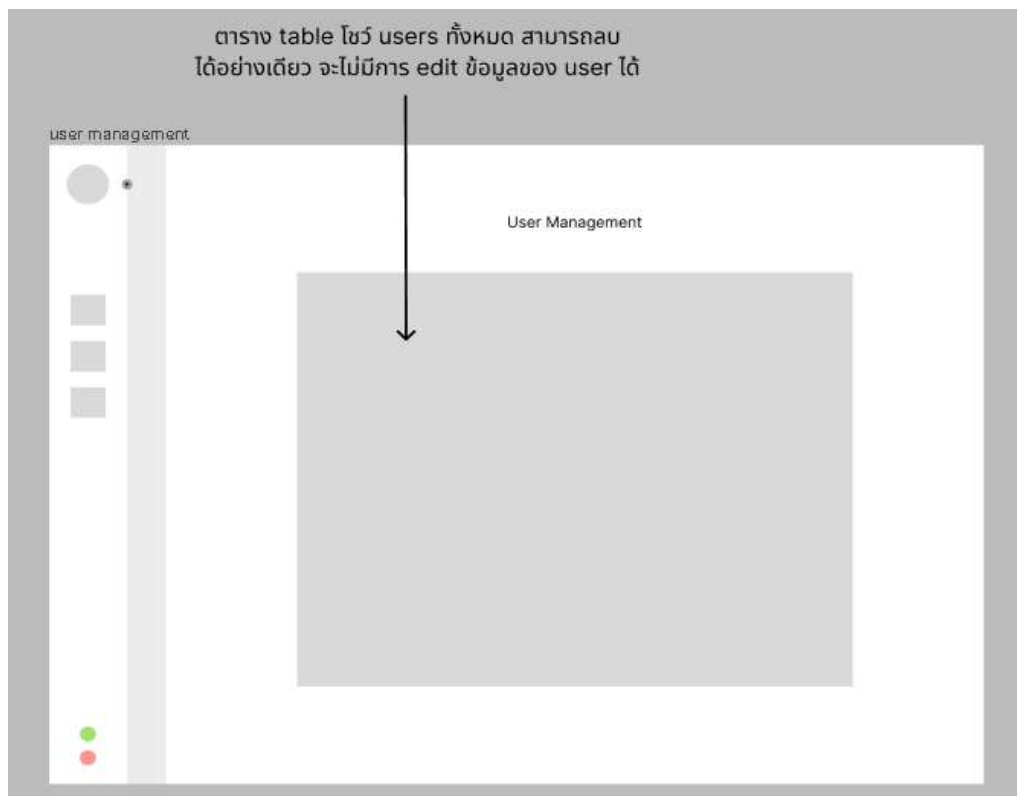


Figure 3.56 Admin interface user management design

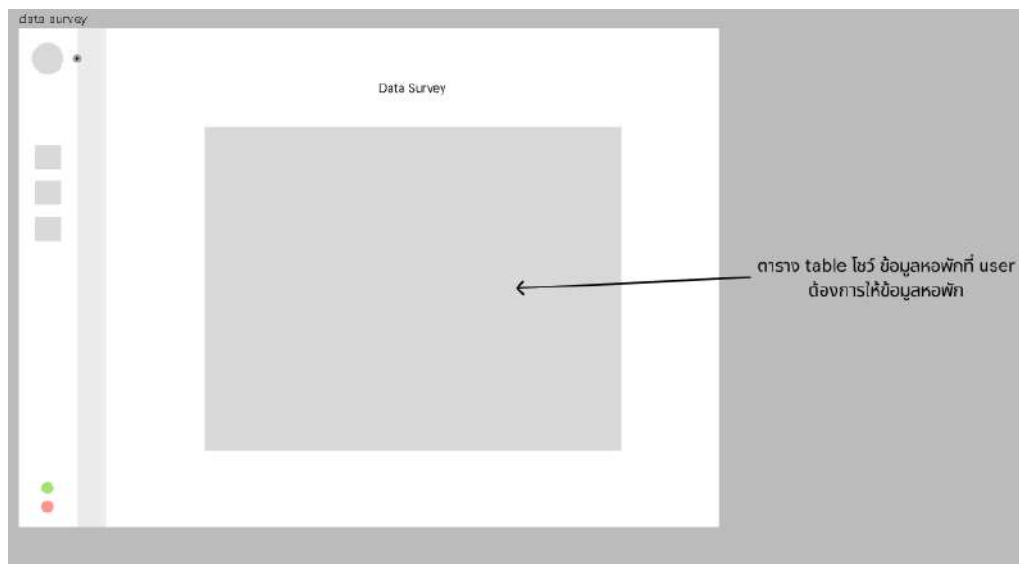


Figure 3.57 Admin interface data survey design

3.3.2 Database Design

3.3.2.1 ER Diagram

The ER diagram depicts the relationships between entities within the `senior` database schema, specifically focusing on the `users` and `provide` tables.

Entities:

1. Users Entity Represents individuals registered within the system.

Attributes include:

`user\_id`: A unique identifier assigned to each user, serving as the primary key.

`username`, `password`: Authentication credentials for user login.

`name`, `lastname`, `email`, `number`, `age`, `birth\_date`, `role`, `sex`: Profile information about users.

2. Provide Entity Represents accommodation details provided by users, derived from machine learning model features.

Attributes include:

`provide\_id`: Unique identifier assigned to each accommodation detail, serving as the primary key.

Various attributes such as `education\_Level`, `security\_deposit\_price`, `facilities`, `distance\_to\_university`, etc., derived from features extracted by the machine learning model.

`user\_id`: A reference to the user_id field in the users table, establishing a connection between accommodations and their respective users.

Figure 3.58 shows the ER diagram

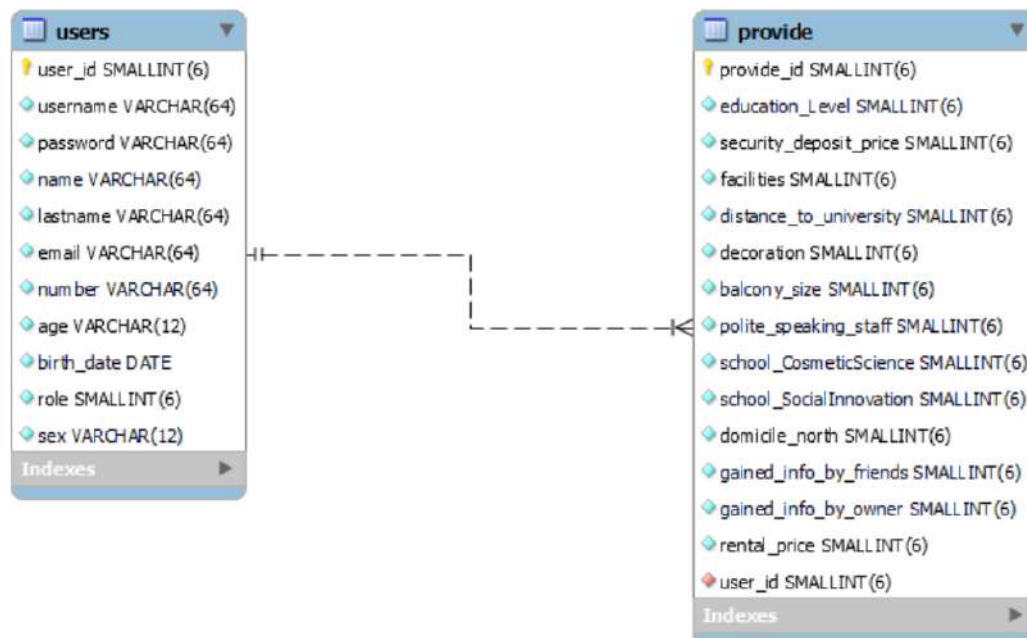


Figure 3.58 ER diagram

3.3.2.2 Database simulation

To facilitate local development and testing of the web application, XAMPP, a cross-platform web server solution, was employed. XAMPP simplifies the setup of a local server environment by bundling Apache, MySQL, PHP, and other essential components required for web development [16]. Figure 3.59 shows the xampp application

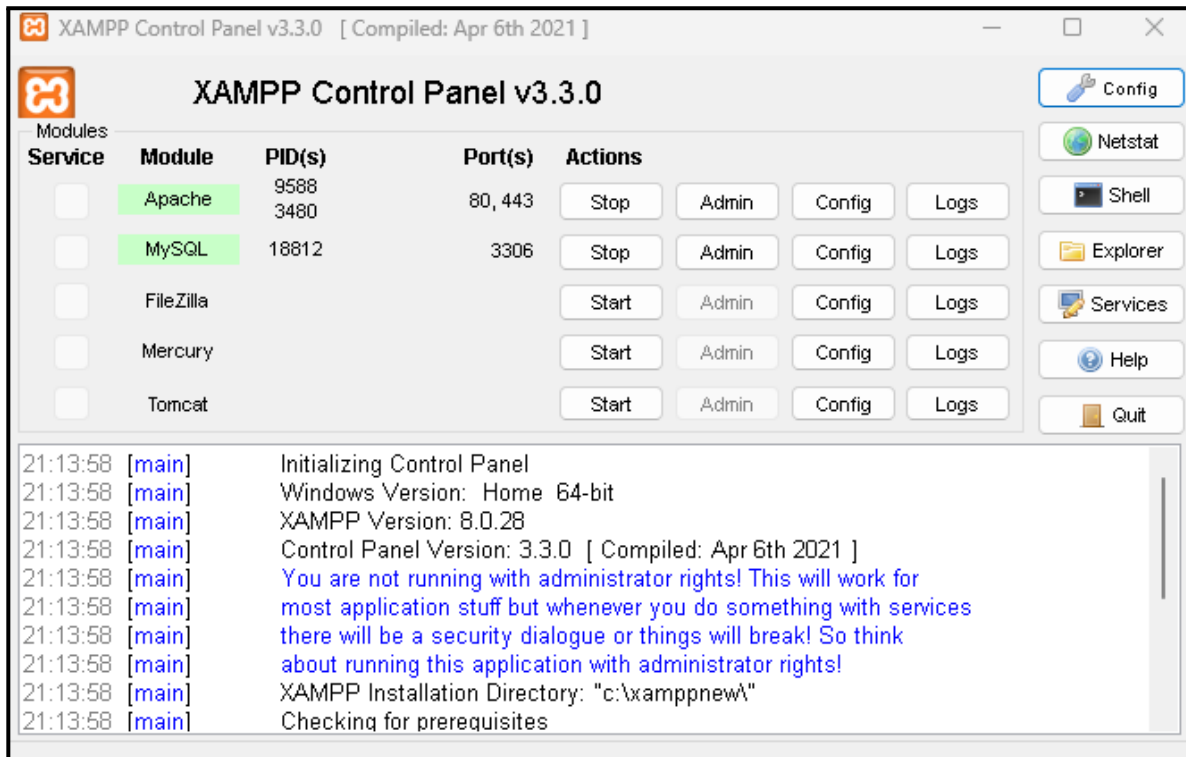


Figure 3.59 Xampp

3.3.3 Element of web application development

3.3.3.1 Front-end development

Front-end development involves creating the user interface (UI) and user experience (UX) components of the web application. The primary technologies used includes.

- 1. HTML (HyperText Markup Language):** HTML provides the structure and content of web pages, defining elements and their layout.
- 2. CSS (Cascading Style Sheets):** CSS is used for styling and presentation, controlling the visual appearance of HTML elements.
- 3. JavaScript (JS):** JS adds interactivity and dynamic behavior to web pages, facilitating user interactions and client-side functionalities.

3.3.3.2 Back-end development

Back-end development focuses on the server-side of the application, handling data processing, logic, and interactions with the database. The technologies utilized encompass

- 1. PHP:** PHP is a server-side scripting language used for generating dynamic content, managing sessions, and handling server requests.
- 2. Python with Flask Framework:** Flask [17] is a micro web framework in Python used for building web applications. In this context, Python is employed to create an API that serves the machine learning model.
- 3. SQL (Structured Query Language):** SQL is used for managing and querying databases. It allows for efficient retrieval and manipulation of data stored in the database.

3.3.3.3 Description of each pages

In our web application will separate into 2 part that is :

1. User Interface

In the user interface, users can register, log in, and predict the price range of accommodations. It includes four key pages:

1.1 Login Page: Users must log in using their registered username and password before accessing our web application.

1.2 Registration Page: Every user is required to register on our platform as certain information is necessary for the prediction function.

1.3 Home Page: The starting point of our web application, guiding users through its functionalities.

1.4 Current Accommodation Form (Optional) Page: This optional page gathers information about the user's current accommodation. The data collected here is stored in our database, contributing to further development of the machine learning model.

1.5 Prediction Page: This page predicts the user-input options. For users who have completed the information on the current accommodation page, an auto-fill function is available to streamline form completion. Once the form is submitted, the prediction results are displayed below the submit button.

2. Admin Interface

2.1 Login page : In our web application we used the same login page of the user interface.

2.2 Home page : The starting point of our admin page, guiding admin through its functionalities.

2.3 User management : This section displays all user information registered on our web application. Additionally, it offers the functionality to remove users from our database.

2.4 Data survey : This section presents all data gathered from the Current Accommodation Form (Optional) Page.

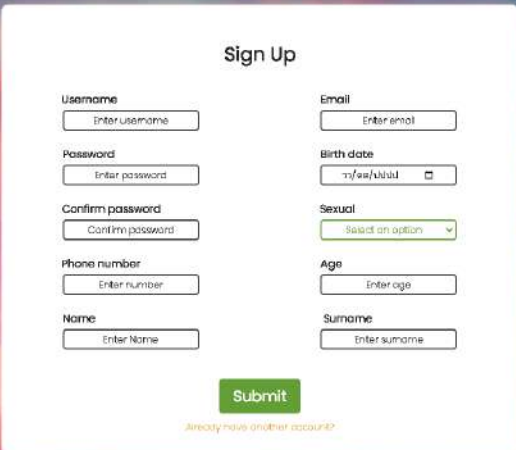
3.3.4 Web application development

The web application development is illustrated in the following figures: Figure 3.60 to Figure 3.67. These figures represent the web application development of user interface and admin interface.



A sign-in form titled "Sign In" is centered on a white background. It features two input fields: "Username" with a placeholder "Enter username" and "Password" with a placeholder "Enter password". Below the password field is a link that says "Don't have an account?". At the bottom of the form is a green "Submit" button.

Figure 3.60 User interface login page



A sign-up form titled "Sign Up" is centered on a white background. It features two columns of input fields. The left column includes: "Username" (placeholder "Enter username"), "Password" (placeholder "Enter password"), "Confirm password" (placeholder "Confirm password"), "Phone number" (placeholder "Enter number"), and "Name" (placeholder "Enter Name"). The right column includes: "Email" (placeholder "Enter email"), "Birth date" (placeholder "mm/dd/yyyy" with a calendar icon), "Sexual" (a dropdown menu with "Select an option"), "Age" (placeholder "Enter age"), and "Surname" (placeholder "Enter surname"). At the bottom of the form is a green "Submit" button and a link that says "Already have an other account?".

Figure 3.61 User interface register page

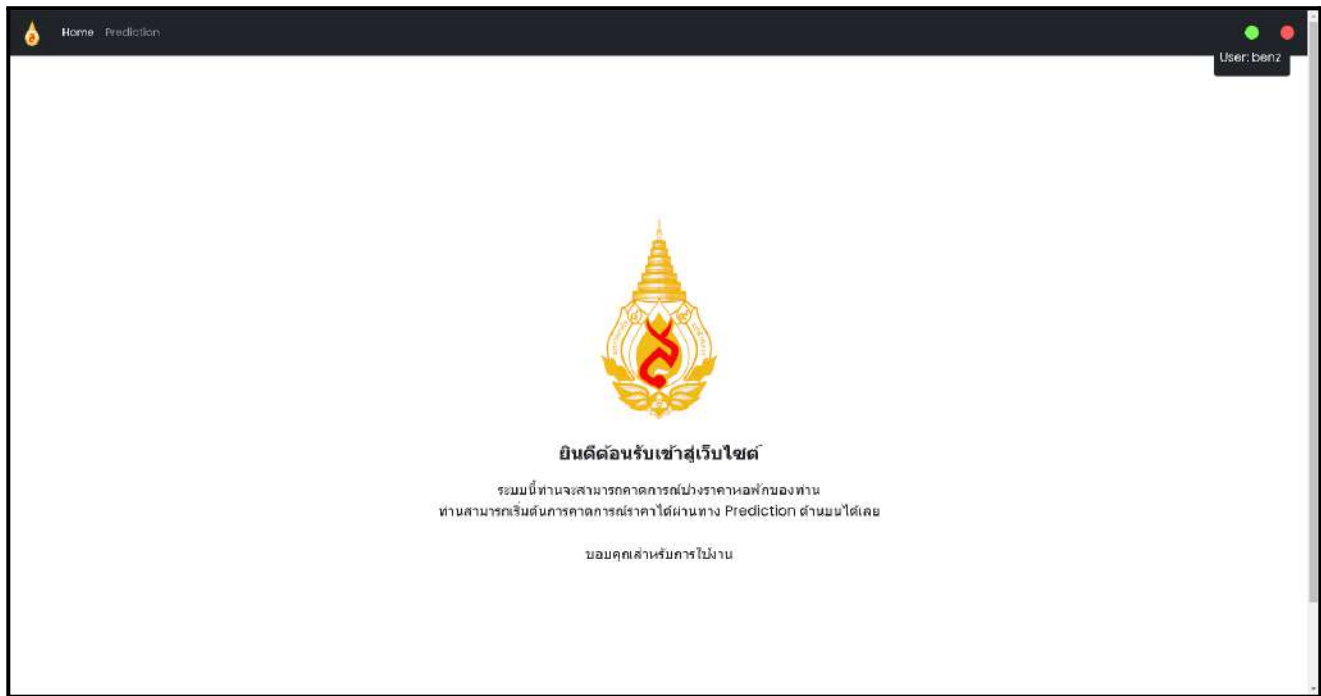


Figure 3.62 User interface home page

Your currently accommodation features(Optional)

Education level	Politeness of officials
Select an option	Select an option
Faculty	Cleanliness around the accommodation area
Select an option	Select an option
Insurance cost	Facilities
Select an option	Select an option
Distance between accommodation and MFU	Domicile
Select an option	Select an option
Decoration	Obtained info
Select an option	Select an option
Balcony size	rental price rate
Select an option	Select an option

Next

Figure 3.63 User interface currently accommodation form (optional) page

 Home Prediction

Accommodation price range prediction

Education level	Polliteness of officials
Select an option	Select an option
Faculty	Cleanliness around the accommodation area
Select an option	Select an option
Insurance cost	Facilities
Select an option	Select an option
Distance between accommodation and MFU	Domicile
Select an option	Select an option
Decoration	Gained info
Select an option	Select an option
Balcony size	
Select an option	

Figure 3.64 User interface prediction page

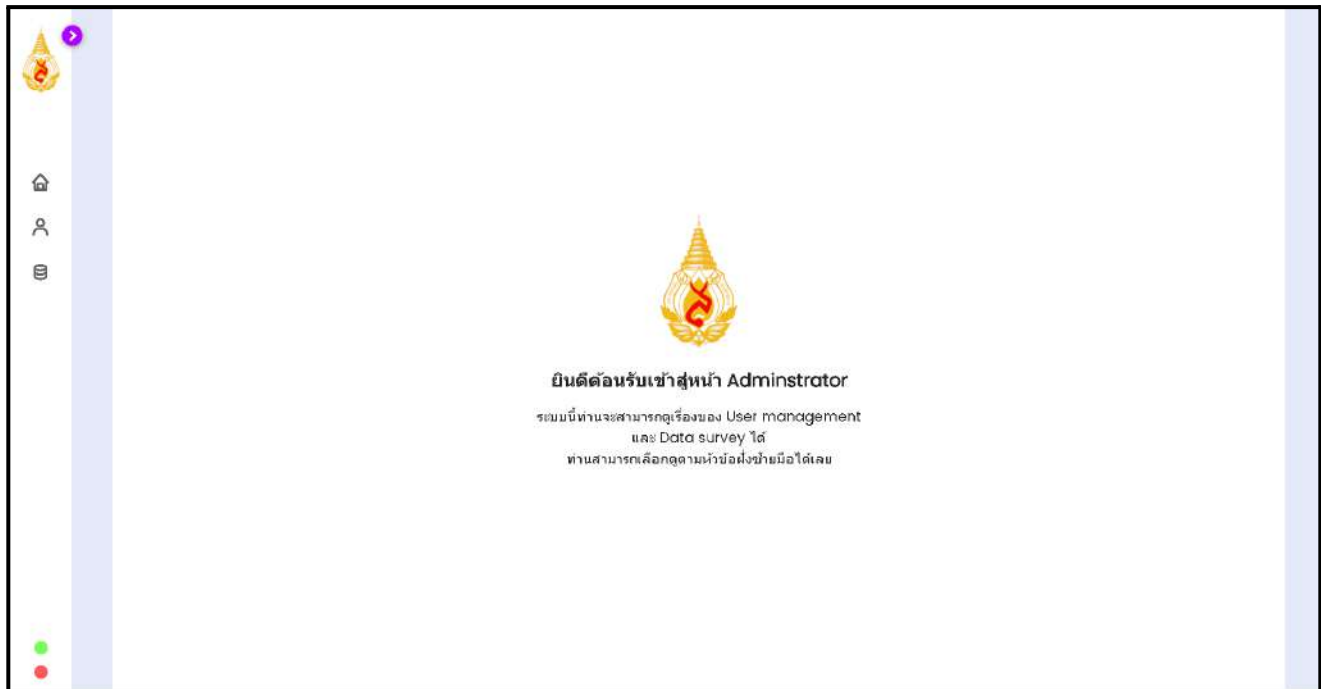


Figure 3.65 Admin interface home page

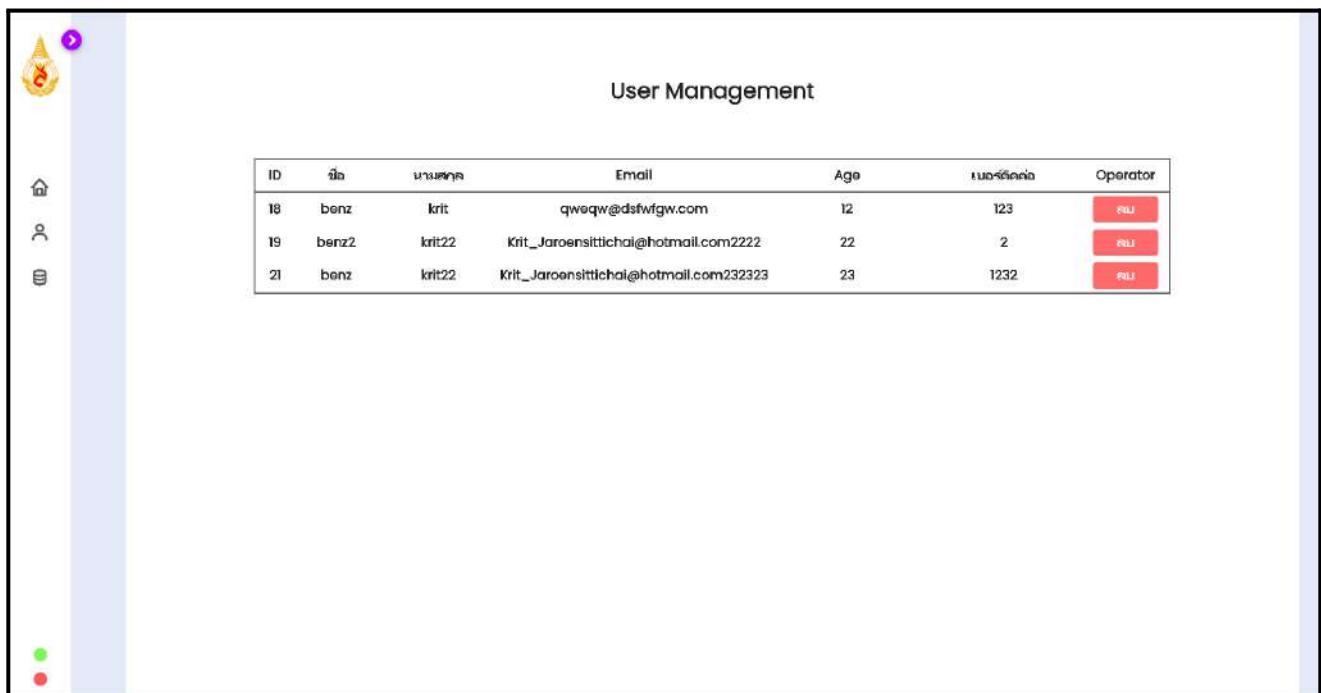
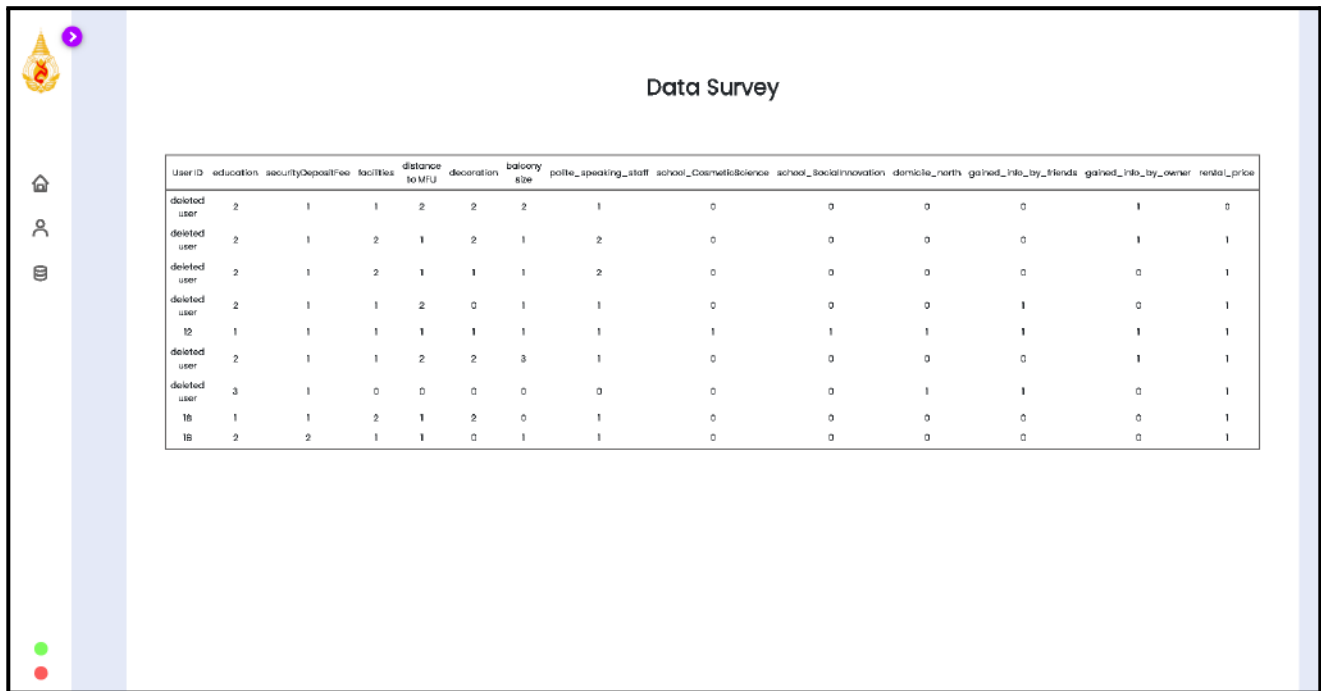


Figure 3.66 Admin interface user management page



The image shows a screenshot of an admin interface titled "Data Survey". On the left, there is a vertical sidebar with a logo at the top, followed by icons for home, user, and a list. At the bottom of the sidebar are two colored circles, green and red. The main content area displays a table with 14 columns and 12 rows of data.

User ID	education	securityDepositFree	facilities	distance to M/FU	decoration	balcony size	potite_speaking_staff	school_CosmeticScience	school_SocialInnovation	domicile_north	gained_info_by_friends	gained_info_by_owner	rental_price
deleted user	2	1	1	2	2	2	1	0	0	0	0	1	0
deleted user	2	1	2	1	2	1	2	0	0	0	0	1	1
deleted user	2	1	2	1	1	1	2	0	0	0	0	0	1
deleted user	2	1	1	2	0	1	1	0	0	0	1	0	1
tb	1	1	1	1	1	1	1	1	1	1	1	1	1
deleted user	2	1	1	2	2	3	1	0	0	0	0	1	1
deleted user	3	1	0	0	0	0	0	0	0	1	1	0	1
tb	1	1	2	1	2	0	1	0	0	0	0	0	1
tb	2	2	1	1	0	1	1	0	0	0	0	0	1

Figure 3.67 Admin interface data survey page

3.3.5 Deploying machine learning model to web application

Deploying a machine learning model to a web application involves integrating the model into the backend using Flask, enabling the application to make predictions based on user inputs

3.3.5.1 Loading machine learning model

Loads a pre-trained machine learning model (model_13_features.joblib) using joblib [18]. Figure 3.68 shows loading machine learning model method

```
import joblib
# Load the model from the file
model = joblib.load('model/model_13_features.joblib')
```

Figure 3.68 Loading machine learning model

3.3.5.2 Defining the Flask App

1. Creates a Flask app instance.
2. Implements CORS (Cross-Origin Resource Sharing) to handle requests from different origins.

Figure 3.98 shows Defining the flask app method

```
from flask_cors import CORS
app = Flask(__name__)
CORS(app)
```

Figure 3.69 Defining the Flask App

3.3.5.3 Prediction Endpoint ('/predict')

1. Listens for POST requests at the /predict endpoint.
2. Receives input data in JSON format via POST requests.
3. Converts the received JSON data into a NumPy array for prediction.
4. Uses the loaded model to generate predictions based on the input features.
5. Returns the prediction results as a JSON response.

Figure 3.70 shows the prediction endpoint process

```
@app.route('/predict', methods=['POST'])
def predict():
    try:
        data = request.get_json()
        print(data)

        # Ensure that data is a dictionary
        if not isinstance(data, dict):
            return jsonify({"error": "Invalid input data"})

        # Convert input data to a NumPy array
        features = np.array([int(value) for value in data.values()]).reshape(1, -1)

        # Make predictions using your machine learning model
        prediction = model.predict(features)

        # Create a response JSON with a 'status' and 'prediction' field
        response = {
            'status': 'success',
            'prediction': prediction.tolist() # Convert predictions to a list
        }

        # Return the prediction as JSON
        return jsonify(response)
```

Figure 3.70 Prediction Endpoint

3.3.5.4 Exception Handling

Includes error handling to catch exceptions and return error responses if any issues occur during prediction. Figure 3.71 shows the exception handling

```
except Exception as e:
    print(str(e))
    # Handle exceptions by returning an error response
    return jsonify({"error": str(e), 'status': 'failed'})
```

Figure 3.71 Exception Handling

3.3.5.5 Run the Application

Starts the Flask app in debug mode. Figure 3.72 shows how to run the application method and Figure 3.73 shows the terminal status when running the application.

```
if __name__ == "__main__":
    app.run(debug=True)
```

Figure 3.72 Run the Application

```
PS F:\modelAPI> python app.py
E:\conda\Lib\site-packages\sklearn\base.py:299: UserWarning: Trying to unpickle estimator LogisticRegression from versio
n 1.3.1 when using version 1.2.1. This might lead to breaking code or invalid results. Use at your own risk. For more in
fo please refer to:
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
  warnings.warn(
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with watchdog (windowsapi)
E:\conda\Lib\site-packages\sklearn\base.py:299: UserWarning: Trying to unpickle estimator LogisticRegression from versio
n 1.3.1 when using version 1.2.1. This might lead to breaking code or invalid results. Use at your own risk. For more in
fo please refer to:
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
  warnings.warn(
* Debugger is active!
* Debugger PIN: 303-271-897
|
```

Figure 3.73 Terminal status

3.3.6 Prediction testing

After we combine the machine learning model with a web application we test the predict function that works correctly that meets our expectation by completing the options. Figure 3.74 shows the prediction testing

The screenshot shows a web application titled "Accommodation price range prediction". It features a form with 12 dropdown menus arranged in two columns. The left column includes: Education level (selected: 1), Faculty (selected: สำนักวิชาเทคโนโลยีการ), Insurance cost (selected: น้อยกว่า 2,000 บาท), Distance between accommodation and MRU (selected: น้อยกว่า 1 กิโลเมตร), Decoration (selected: ไม่มีสิ่งใด), and Balcony size (selected: ไม่มีสิ่งใด). The right column includes: Politeness of officials (selected: ไม่มีสิ่งใด), Cleanliness around the accommodation area (selected: ไม่มีสิ่งใด), Facilities (selected: ไม่มีสิ่งใด), Dormitory (selected: ไม่มีสิ่งใด), and Gained info (selected: ไม่มีสิ่งใด). Below the form are two buttons: "Previous Page" (orange) and "Submit" (green). The result displayed at the bottom is: "ผลการประเมินช่วงราคาห้องพักของคุณคือ น้อยกว่า 2000 บาท".

Figure 3.74 Prediction testing

CHAPTER 4

RESULTS

4.1 Performance Comparison of Machine Learning Algorithms

The performance of multiple machine learning algorithms was evaluated using various evaluation metrics to determine their efficacy in predicting expected dormitory price ranges around Mae Fah Luang University (MFU) campus. The following table presents a comparative analysis of the performance scores obtained from each algorithm

4.1.1 Default model (None of optimization)

The default model employed standard parameters and features, utilizing a total of 70 default features for training purposes as can be seen in table 4.1

		Model			
		Class	random	decision	logistic
Default	Precision	0	1	0.33	0.2
		1	0.8	0.78	0.84
		2	0.78	0.52	0.78
	Recall	0	0.14	0.43	0.14
		1	0.95	0.77	0.9
		2	0.54	0.5	0.69
		Avg recall	0.54	0.57	0.58
	F1	0	0.25	0.38	0.17
		1	0.87	0.77	0.87
		2	0.64	0.51	0.73
	Accuracy		0.8	0.68	0.8

Table 4.1 Hyperparameter tuning table

4.1.2 Hyperparameter tuning(Grid search)

This section delves into the performance metrics of machine learning models after fine-tuning their hyperparameters using grid search techniques [15] as can be seen in table 4.2

		Model			
		Class	random	decision	logistic
Hyperparameter	Precision	0	0.5	0.25	0.29
		1	0.78	0.78	0.85
		2	0.73	0.54	0.75
	Recall	0	0.14	0.29	0.29
		1	0.94	0.79	0.87
		2	0.42	0.5	0.69
		Avg recall	0.5	0.53	0.62
	F1	0	0.22	0.27	0.29
		1	0.85	0.79	0.86
		2	0.54	0.52	0.72
	Accuracy		0.77	0.69	0.79

Table 4.2 Hyperparameter tuning table

4.1.3 Resampling technique

This section presents the performance comparison of machine learning models after applying resampling techniques to address class imbalance issues as can be seen in table 4.3

			Model		
		Class	random	decision	logistic
Resampling	Precision	0	0.5	0.25	0.21
		1	0.86	0.78	0.85
		2	0.69	0.42	0.57
	Recall	0	0.29	0.43	0.43
		1	0.9	0.68	0.68
		2	0.69	0.5	0.77
		Avg recall	0.63	0.54	0.63
	F1	0	0.36	0.32	0.29
		1	0.88	0.73	0.76
		2	0.69	0.46	0.66
	Accuracy		0.81	0.62	0.68

Table 4.3 Resampling table

4.1.4 Sequential features selection

This section presents the model performance based on varying numbers of selected features using sequential feature selection techniques. The sequential features selection 13 features to 40 features is illustrated in the following tables: Table 4.4 to Table 4.14. These tables represent the sequential features selection of scores that is precision, recall, f1, average recall and accuracy of each model with different algorithms.

			Model with resampling		
		Class	random	decision	logistic
13	Precision	0	0.22	0.21	0.3
		1	0.87	0.81	0.93
		2	0.62	0.57	0.73
	Recall	0	0.57	0.57	0.86
		1	0.68	0.67	0.73
		2	0.77	0.62	0.85
		Avg recall	0.67	0.62	0.81
	F1	0	0.32	0.31	0.44
		1	0.76	0.73	0.82
		2	0.69	0.59	0.79
	Accuracy		0.69	0.65	0.77

Table 4.4 13 selected features

			Model with resampling		
		Class	random	decision	logistic
15	Precision	0	0.25	0.3	0.24
		1	0.85	0.9	0.93
		2	0.73	0.67	0.7
	Recall	0	0.14	0.86	0.71
		1	0.88	0.71	0.68
		2	0.73	0.77	0.88
		Avg recall	0.58	0.78	0.76
	F1	0	0.18	0.44	0.36
		1	0.87	0.79	0.79
		2	0.73	0.71	0.78
	Accuracy		0.8	0.73	0.73

Table 4.5 15 selected features

			Model with resampling		
		Class	random	decision	logistic
18	Precision	0	0.33	0.38	0.25
		1	0.87	0.84	0.92
		2	0.63	0.54	0.71
	Recall	0	0.57	0.71	0.71
		1	0.77	0.76	0.71
		2	0.73	0.58	0.85
		Avg recall	0.69	0.68	0.76
	F1	0	0.42	0.5	0.37
		1	0.82	0.8	0.8
		2	0.68	0.56	0.77
	Accuracy		0.75	0.71	0.74

Table 4.6 18 selected features

			Model with resampling		
		Class	random	decision	logistic
20	Precision	0	0.08	0.2	0.17
		1	0.83	0.86	0.9
		2	0.82	0.62	0.67
	Recall	0	0.14	0.43	0.43
		1	0.81	0.71	0.69
		2	0.69	0.77	0.85
		Avg recall	0.55	0.64	0.66
	F1	0	0.1	0.27	0.24
		1	0.82	0.77	0.78
		2	0.75	0.69	0.75
	Accuracy		0.74	0.7	0.71

Table 4.7 20 selected features

			Model with resampling		
		Class	random	decision	logistic
23	Precision	0	0.23	0.29	0.28
		1	0.88	0.88	0.92
		2	0.83	0.76	0.79
	Recall	0	0.43	0.71	0.71
		1	0.83	0.78	0.77
		2	0.77	0.73	0.85
		Avg recall	0.68	0.74	0.78
	F1	0	0.3	0.42	0.4
		1	0.86	0.83	0.84
		2	0.8	0.75	0.81
	Accuracy		0.79	0.77	0.78

Table 4.8 23 selected features

			Model with resampling		
		Class	random	decision	logistic
25	Precision	0	0.2	0.33	0.28
		1	0.85	0.89	0.92
		2	0.78	0.76	0.79
	Recall	0	0.29	0.71	0.71
		1	0.85	0.81	0.77
		2	0.69	0.73	0.85
		Avg recall	0.61	0.75	0.78
	F1	0	0.24	0.45	0.4
		1	0.85	0.85	0.84
		2	0.73	0.75	0.81
	Accuracy		0.77	0.78	0.78

Table 4.9 25 selected features

			Model with resampling		
		Class	random	decision	logistic
28	Precision	0	0.31	0.25	0.28
		1	0.89	0.87	0.92
		2	0.76	0.69	0.79
	Recall	0	0.71	0.57	0.71
		1	0.79	0.77	0.77
		2	0.73	0.69	0.85
		Avg recall	0.74	0.68	0.78
	F1	0	0.43	0.35	0.4
		1	0.84	0.82	0.84
		2	0.75	0.69	0.81
	Accuracy		0.77	0.74	0.78

Table 4.10 28 selected features

			Model with resampling		
		Class	random	decision	logistic
30	Precision	0	0.2	0.18	0.32
		1	0.85	0.84	0.94
		2	0.74	0.65	0.79
	Recall	0	0.29	0.43	0.86
		1	0.85	0.73	0.77
		2	0.65	0.65	0.85
		Avg recall	0.60	0.60	0.83
	F1	0	0.24	0.25	0.46
		1	0.85	0.78	0.85
		2	0.69	0.65	0.81
	Accuracy		0.77	0.69	0.79

Table 4.11 30 selected features

			Model with resampling		
		Class	random	decision	logistic
35	Precision	0	0.36	0.18	0.26
		1	0.9	0.87	0.92
		2	0.8	0.7	0.79
	Recall	0	0.71	0.43	0.71
		1	0.83	0.74	0.76
		2	0.77	0.73	0.85
		Avg recall	0.77	0.63	0.77
	F1	0	0.48	0.25	0.38
		1	0.87	0.8	0.83
		2	0.78	0.72	0.81
	Accuracy		0.81	0.72	0.77

Table 4.12 35 selected features

			Model with resampling		
		Class	random	decision	logistic
38	Precision	0	0.31	0.25	0.28
		1	0.88	0.87	0.92
		2	0.9	0.69	0.79
	Recall	0	0.57	0.57	0.71
		1	0.87	0.77	0.77
		2	0.73	0.69	0.85
		Avg recall	0.72	0.68	0.78
	F1	0	0.4	0.35	0.4
		1	0.88	0.82	0.84
		2	0.81	0.69	0.81
	Accuracy		0.82	0.74	0.78

Table 4.13 38 selected features

			Model with resampling		
		Class	random	decision	logistic
40	Precision	0	0.27	0.29	0.26
		1	0.89	0.88	0.92
		2	0.87	0.76	0.79
	Recall	0	0.57	0.71	0.71
		1	0.83	0.78	0.76
		2	0.77	0.73	0.85
		Avg recall	0.72	0.74	0.77
	F1	0	0.36	0.42	0.38
		1	0.86	0.83	0.83
		2	0.82	0.75	0.81
	Accuracy		0.8	0.77	0.77

Table 4.14 40 selected features

4.1.5 Evaluate results comparison

The scores achieved through Sequential Feature Selection, encompassing models optimized via Hyperparameter (Grid Search) and Resampling techniques, revealed significant variations across different feature sets. Notably, models utilizing logistic regression algorithm 13 features that have got 0.81 of average recall and 35 features that have got 0.83 of average recall stood out due to their higher performance metrics. Considering the objective of deploying a machine learning model into a user-centric web application, the model employing 13 features was selected. This decision was rooted in the pursuit of balancing predictive accuracy and user convenience, aligning with the aim of ensuring an expected user experience in the web application.

CHAPTER 5

CONCLUSION

5.1 Conclusion

This project aimed to predict accommodation prices range around Mae Fah Luang University and develop a user-friendly web application. The dataset revealed an imbalance across all classes, with Class 3 particularly low due to outliers and scarce records. However, the handling of this imbalance led to improved model accuracy.

Strategic measures like resampling, hyperparameter tuning, and feature selection significantly enhanced model performance. Two standout models emerged: Logistic (13 features) achieved an average recall of 0.81, while Logistic (35 features) scored 0.85. Considering user-friendliness, the Logistic model with 13 features was prioritized for the web application.

5.2 Discussion

Addressing the dataset's imbalance involved the exclusion of Class 3 due to minimal data and numerous outliers, significantly impacting model inclusivity. Nevertheless, this step fortified the model's robustness. Choosing the simpler model for web deployment sought to optimize user interaction and accessibility.

This project showcased the successful integration of machine learning and web development, allowing seamless deployment of a predictive model within a user-centric web interface. Future endeavors could explore methods to enhance model inclusivity without compromising performance by potentially expanding the dataset.

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APPENDIX

APPENDIX

SOURCE CODE

First, an example of an imbalance script.

```
for i in ['13','15','18','20','23','25','28','30','35','38','40']:
    file_path = f'../data/split/data_features_selected_{i}.train.csv'
    model_path = f'model/model_6_11_23/model_logistic_selected_{i}_cross_validation.joblib'
    data_train = pandas.read_csv(file_path).set_index('id')
    data_class_0 = data_train[data_train['rental_price'] != 0]
    data_class_1 = data_train[data_train['rental_price'] != 1]
    desired_samples_per_class = 124
    under = NearMiss(sampling_strategy={1: desired_samples_per_class,2:62})
    over = SMOTE(sampling_strategy={0: 62,2:62})

    # Check if there are multiple classes in data_class_0 and data_class_1
    if data_class_1['rental_price'].nunique() != 1:
        # Apply oversampling to class 0 only if there are multiple classes
        X_train_resampled_over, y_train_resampled_over = over.fit_resample(data_class_1.drop(['rental_price'],
axis='columns'), data_class_1['rental_price'])

    if data_class_0['rental_price'].nunique() != 0 :
        # Apply undersampling to class 1 only if there are multiple classes
        X_train_resampled_under, y_train_resampled_under = under.fit_resample(data_class_0.drop(['rental_price'],
axis='columns'), data_class_0['rental_price'])

    # Combine the resampled data
    if 'X_train_resampled_over' in locals() and 'X_train_resampled_under' in locals():
        X_train_resampled = pandas.concat([X_train_resampled_over, X_train_resampled_under])
        y_train_resampled = pandas.concat([y_train_resampled_over, y_train_resampled_under])
    elif 'X_train_resampled_over' in locals():
        X_train_resampled = X_train_resampled_over
        y_train_resampled = y_train_resampled_over
    elif 'X_train_resampled_under' in locals():
        X_train_resampled = X_train_resampled_under
        y_train_resampled = y_train_resampled_under
    else:
        # If no resampling was performed, use the original data
```

```

X_train_resampled = data_train.drop(['rental_price'], axis='columns')
y_train_resampled = data_train['rental_price']
print('-----value counts ',i,'-----\n',y_train_resampled.value_counts(),)
model.fit(
X=X_train_resampled,
y= y_train_resampled
)
cv_scores = cross_val_score(RandomForestClassifier(n_estimators=40),X_train_resampled,
y_train_resampled,cv=10)
joblib.dump(model,model_path)
recall_scores = cross_val_score(model, X_train_resampled, y_train_resampled, cv=5, scoring='recall_macro')

print(i+' features\nMean CV Accuracy Score:', cv_scores.mean(),'\n-----')
print("Mean CV Recall Score:", recall_scores.mean())

```

Second, an example of a model API file.

```

from flask import Flask, request, jsonify
from flask_cors import CORS
import numpy as np
import joblib
# Load the model from the file
model = joblib.load('model/model_13_features.joblib')

app = Flask(__name__)
CORS(app)

@app.route('/predict', methods=['POST'])
def predict():
    try:
        data = request.get_json()
        print(data)

        # Ensure that data is a dictionary
        if not isinstance(data, dict):
            return jsonify({"error": "Invalid input data"})

        # Convert input data to a NumPy array
        features = np.array([int(value) for value in data.values()]).reshape(1, -1)

```

```

# Make predictions using your machine learning model
prediction = model.predict(features)

# Create a response JSON with a 'status' and 'prediction' field
response = {
    'status': 'success',
    'prediction': prediction.tolist() # Convert predictions to a list
}

# Return the prediction as JSON
return jsonify(response)
except Exception as e:
    print(str(e))
    # Handle exceptions by returning an error response
    return jsonify({"error": str(e), 'status': 'failed'})

if __name__ == "__main__":
    app.run(debug=True)

```